# $t\bar{t}$ and $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$: samples, the ten angles, and the $t\bar{t}$ density $f_{t\bar{t}}$ by OSDE as a signal-agnostic discriminant

Internal note

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**Abstract**

The orthogonal-series density $f_{t\bar{t}}$ of the ten $t\bar{t}$ angles, fitted on $t\bar{t}$ only, separates $hh \to b\bar{b}\,\ell\nu\,q\bar{q}'$ from $t\bar{t}$ at parton level (AUC 0.67) but not after reconstruction (0.53 at particle level, 0.51 at detector level). The loss is traced to the $t\bar{t}$-hypothesis reconstruction, which rebuilds the off-shell $hh$ events as on-shell $t\bar{t}$; the information survives in the resonance masses. Multiplying $f_{t\bar{t}}$ by one Breit–Wigner factor per resonance mass, with no fitted coefficients, recovers the parton-level performance at particle level (0.69) and most of it at detector level (0.60). Only about 20 angular terms matter; realistic quark tagging changes nothing once the masses are in. Comparisons with other unsupervised and with supervised machine-learning methods, a BSM benchmark and a pileup study are included.

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# 1 Introduction and summary

This note documents a study of the orthogonal-series density estimate (OSDE) $f_{t\bar{t}}(\mathbf{x})$ of the ten $t\bar{t}$ decay and production angles $\mathbf{x} = \{Y, \Theta, \theta, \phi, \theta', \phi', \tilde{\theta}, \tilde{\phi}, \tilde{\theta}', \tilde{\phi}'\}$ defined in the paper, as a signal-agnostic (unsupervised) discriminant: $y = f_{t\bar{t}}(\mathbf{x})$ is low for events that do not look like $t\bar{t}$, whatever they are. The test signal is $hh \to b\bar{b}\,WW^* \to b\bar{b}\,\ell\nu\,q\bar{q}'$, which has the same final state. The study

has four parts:

1. Monte Carlo samples of $t\bar{t}$ and $hh$ at parton level (MadGraph5), particle level (Pythia8) and detector level (Delphes, ATLAS card, no pileup), with the full $t\bar{t}$ reconstruction, and the ten angles computed at each level (Sections 2–3).
2. The density $f_{t\bar{t}}$ by OSDE in the paper's basis, its validation, the number of coefficients that matter, the orders needed at particle and detector level, and a comparison with a spherical-harmonic basis (Sections 4–7).
3. How well $f_{t\bar{t}}$ separates $hh$ from $t\bar{t}$ compared with other unsupervised methods and with supervised machine learning on the same inputs, and why the separation seen at parton level disappears after reconstruction (Sections 9–10).
4. The fix: putting the momenta back through the resonance masses, which recovers the parton-level performance at particle and detector level, and the effect of realistic quark tagging (Section 11). A BSM benchmark and a pileup study are in Sections 12 and 13.

The headline answers are collected in Table 1. Throughout, AUC is the area under the ROC curve for $hh$ versus $t\bar{t}$ (0.5 = no separation), always evaluated on events that were not used in any fit: the independent $t\bar{t}$ sample, or the fit sample scored leave-one-out where stated.

**Table 1:** Headline answers.

| question | answer |
|---|---|
| Does $f_{t\bar{t}}$ separate $hh$ from $t\bar{t}$ at parton level? | Yes: AUC 0.665 (0.685 with the lepton trigger), using no $hh$ information |
| Is it the best unsupervised method? | Narrowly: $+0.011 \pm 0.003$ over a normalizing flow; well ahead of a Gaussian mixture and an autoencoder |
| How far from the best possible with these angles? | A supervised classifier on the same angles reaches 0.89 |
| Does $y$ help a supervised classifier? | No (change within $\pm 0.002$) |
| What happens at gen and reco level? | AUC drops to 0.53 and 0.51 |
| Why? | The $\chi^2$ fit and the $m_W$-constrained neutrino rebuild $hh$ events as $t\bar{t}$; the information survives in observables that do not assume $t\bar{t}$ (BDT AUC 0.87) |
| How many coefficients matter? | Cross-validation keeps 326, 385, 459, 342 of 14 400 (parton, parton+trig, gen, reco); ∼20 terms do as well as all of them |
| Can the momenta be put back? | Yes: one Breit–Wigner factor per resonance mass. Parton 0.65→1.00, gen 0.53→0.69, reco 0.51→0.60 |

The slides (`slides/slides.pdf`) show the same material in the same order; the section titles here match the slide sections. All numbers in this note are read from the analysis outputs by `note/make_note.py`, so the note and the slides cannot drift apart.

## 2 Samples and event levels

### 2.1 Generation

All samples are produced by `generate_samples.py`: MadGraph5_aMC@NLO 3.6.7 at leading order (4-flavour proton, $\sqrt{s} = 13$ TeV), Pythia8 for the parton shower and hadronization, and

Delphes 3.5.1 with the ATLAS card and no pileup. The $t\bar{t}$ process is generated with the full decay chain so that spin correlations are kept and without generator-level cuts: $pp \to t\bar{t}$, $t \to bW^+(\to \ell^+\nu)$, $\bar{t} \to \bar{b}W^-(\to q\bar{q}')$ plus the charge conjugate, with $\ell = e, \mu$. The cross section is $\sigma = 136.5$ pb, i.e. the LO $t\bar{t}$ cross section times $2\,\mathcal{B}(W \to \ell\nu)\,\mathcal{B}(W \to q\bar{q}')$. Two statistically independent $t\bar{t}$ samples are used throughout: the *fit sample* (100k events, seed 1001), on which every density is fitted, and the *independent sample* (10k events, seed 12345), on which every result is evaluated.

The $hh$ sample (20k events) uses MadGraph's built-in `heft` model (effective $ggh$ and $gghh$ couplings in the $m_t \to \infty$ limit): $gg \to hh$, $h \to b\bar{b}$, $h \to \ell\nu q\bar{q}'$ generated as one process. Because $h \to WW^*$ has one off-shell $W$, the $h \to \ell\nu q\bar{q}'$ decay is a four-body decay. Its cross section times branching ratio, 0.112 fb, is the LO HEFT value with MadGraph defaults and not an SM prediction, so all $hh$ distributions are shown at unit area. For $t\bar{t}$-hypothesis quantities $hh$ goes through exactly the same reconstruction; at parton level the pairing of the $b$ quarks with the $W$ bosons is the one closest to the $t\bar{t}$ hypothesis (both $m_{Wb}$ closest to $m_t$).

## 2.2 Event levels and selection

Four event levels are used (Table 2). *Parton* is the MadGraph output with no cuts. *Parton+trig* adds only the offline single-lepton (trigger) cuts to the charged lepton, $p_T > 25$ GeV and $|\eta| < 2.47$ ($e$) or 2.5 ($\mu$), and keeps 71.7% of the events. *Gen* is the particle level: anti-$k_t$ $R = 0.4$ GenJets, the prompt lepton, and the missing momentum from the sum of all neutrinos. Prompt leptons are status-1 $e/\mu$ from a $W$, $Z$ or $h$, or from a $\tau$ that itself comes from one of these (in $h \to WW^*$ the lepton can come directly from the $h$ in the record); leptons from $B$ and $D$ decays, also via $\tau$, are not prompt. GenJets within $\Delta R < 0.4$ of a prompt lepton are removed, since the lepton is clustered into a jet otherwise. A GenJet is $b$-labelled if a $b$ quark with $p_T > 5$ GeV lies within $\Delta R < 0.4$. *Reco* is the Delphes output: jets, electrons, muons and the calorimeter missing transverse momentum, with the Delphes $b$-tag bit. Jets need $p_T > 25$ GeV and $|\eta| < 2.5$; leptons the same cuts as the trigger.

The gen and reco selection requires exactly one lepton, at least four jets and at least one $b$-jet, followed by the $\chi^2$ $t\bar{t}$ reconstruction

$$\chi^2 = \left(\frac{m_{jj} - m_W}{\sigma_W}\right)^2 + \left(\frac{m_{jjb} - m_t}{\sigma_t}\right)^2 + \left(\frac{m_{\ell\nu b} - m_t}{\sigma_t}\right)^2, \tag{1}$$

minimized over all assignments of the six leading jets to $(b_{\mathrm{had}}, b_{\mathrm{lep}}, q_1, q_2)$ with $\min(n_b, 2)$ $b$-tagged jets in the two $b$ slots, and over the two solutions of the neutrino $p_z$ from the $m_W$ constraint, so that the $m_t$ term resolves the two-fold ambiguity ($\sigma_W = 10$, $\sigma_t = 15$ GeV). The cutflow is in Table 3: 41.3% of the $t\bar{t}$ events survive at gen level and 24.4% at reco level; $hh$ leptons are softer (from the $W^*$) and less isolated, so its efficiency is lower. Reco and gen missing momentum agree for muon and electron events (reco minus gen along the lepton direction: $-5.9$ vs $-5.2$ GeV), which checks that the calorimeter missing momentum does not lose the muons.

## 2.3 Reconstruction quality

All four jets are assigned correctly in 23% (gen) and 18% (reco) of the reconstructed $t\bar{t}$ events. The reconstructed masses (Table 4) have broad inclusive peaks that come from wrong jet choices and from quarks outside the acceptance, not from the jet response: with the correct jets the hadronic top and $W$ peaks are narrow.

**Table 2:** Event levels.

| | parton | parton+trig | gen (particle) | reco |
|---|---|---|---|---|
| source | MG5 LHE | MG5 LHE | Pythia8 | Delphes ATLAS |
| selection | none | $\ell$ cuts only | $1\ell, \geq 4j, \geq 1b, \chi^2$ | $1\ell, \geq 4j, \geq 1b, \chi^2$ |
| events kept | 100% | 71.7% | 41.3% | 24.4% |
| $E_T^{\text{miss}}$ | $p_T^\nu$ | $p_T^\nu$ | $\sum$ all $\nu$ | $-\sum$ calo towers |

**Table 3:** Cutflow of the independent $t\bar{t}$ sample and of the $hh$ sample (reco); fractions relative to all generated events.

| selection | gen | | reco | | $hh$ reco | |
|---|---|---|---|---|---|---|
| All events | 10000 | 100.0% | 10000 | 100.0% | 20000 | 100.0% |
| exactly 1 lepton | 7071 | 70.7% | 4907 | 49.1% | 5403 | 27.0% |
| $\geq 4$ jets | 4196 | 42.0% | 2830 | 28.3% | 2133 | 10.7% |
| $\geq 1$ b-jet | 4161 | 41.6% | 2454 | 24.5% | 1887 | 9.4% |
| $\geq 2$ b-jets | 3305 | 33.1% | 1221 | 12.2% | 920 | 4.6% |
| $\chi^2$ reconstruction (from $\geq 1$ b) | 4134 | 41.3% | 2440 | 24.4% | 1878 | 9.4% |

The neutrino $p_z$ (Figure 1) has a 68% half-width of 54 GeV at gen and 72 GeV at reco level, improving to 49 GeV when all jets are correct. The $m_W$ constraint has no real solution in 32% of the events (the discriminant is then set to zero). When two real solutions exist, the top-mass term of the $\chi^2$ picks the one closer to the truth in 63% of the events, which is not better than simply taking the smaller $|p_z|$ (65%).

For $hh$ the reconstruction under the $t\bar{t}$ hypothesis is already visible in the masses: the hadronic boson is the off-shell $W^*$ in 50% of the events, with a true median mass of 32 GeV, but the $\chi^2$ picks jet pairs near $m_W$ (median 95 GeV at gen and 89 GeV at reco level). This is the first sign of the effect studied in Section 10.

## 3 The ten angles

The angles are defined exactly as in the paper (Figure 3). $Y$ is the rapidity of the $t\bar{t}$ system in the laboratory and $\Theta$ the polar angle of the leptonic top in the $t\bar{t}$ rest frame. For each top, $(\theta, \phi)$ are the polar and azimuthal angles of the $W$ in the top rest frame, with $z = \hat{p}_t^{\text{lab}}$, $y \propto \hat{z}_{\text{beam}} \times z$ and $x = y \times z$; and $(\theta', \phi')$ are the angles of the fermion in the $W$ rest frame reached by a standard boost from the top frame, with $z' = \hat{p}_W$, $y' \propto z \times z'$ and $x' = y' \times z'$, i.e. the rotation $D(\phi, \theta, 0)$. On the leptonic side the fermion is the charged lepton; on the hadronic side (tilde angles) it is the down-type quark, which plays the same role in the $V-A$ decay. At gen and reco level the down-type jet is chosen by matching to the parton-level quarks, i.e. perfect up/down tagging as the paper assumes; realistic tagging is studied in Section 11.6. The implementation was checked

**Table 4:** Mass resolution: median ± 68% half-width (GeV).

| | $m_{t,\text{had}}$, all | $m_{t,\text{had}}$, correct jets | $m_{W,\text{had}}$, all | $m_{W,\text{had}}$, correct jets |
|---|---|---|---|---|
| gen | $182 \pm 47$ | $175 \pm 14$ | $85 \pm 29$ | $83 \pm 9$ |
| reco | $177 \pm 44$ | $170 \pm 16$ | $84 \pm 25$ | $80 \pm 10$ |

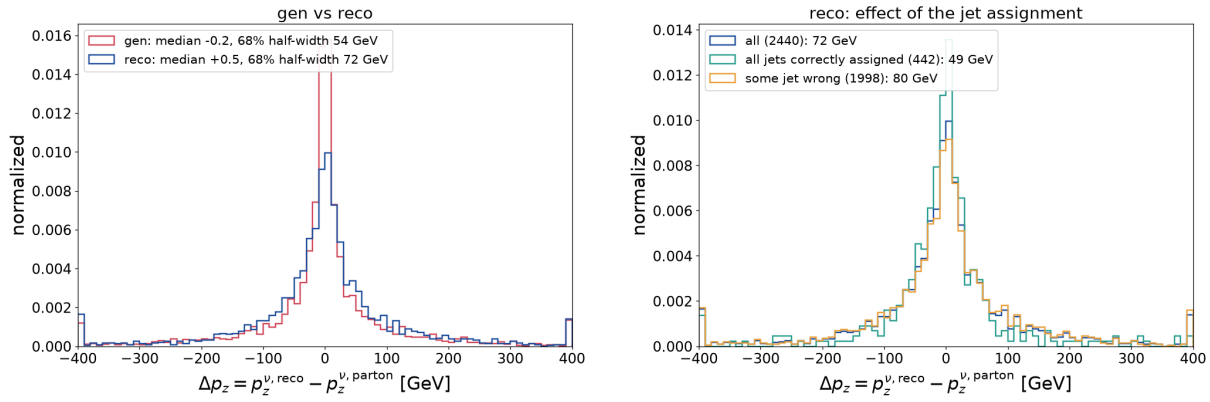



**Figure 1:** Neutrino $\Delta p_z = p_z^{\nu,\mathrm{reco}} - p_z^{\nu,\mathrm{parton}}$ at gen and reco level (left) and at reco level split by the correctness of the jet assignment (right).

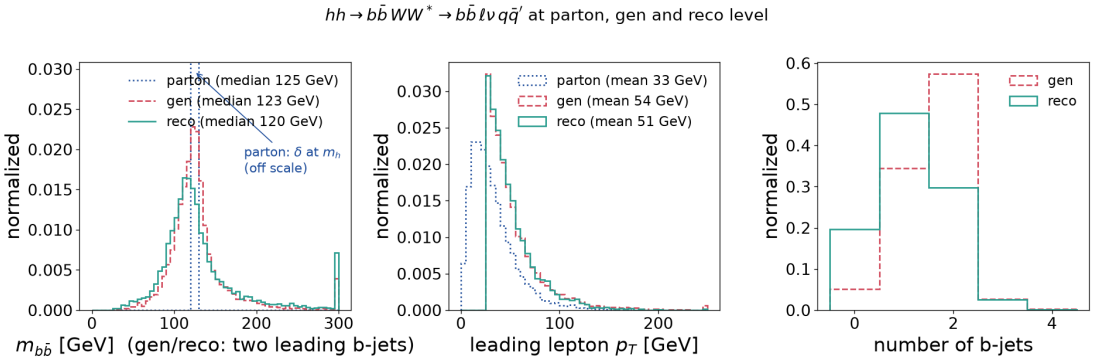
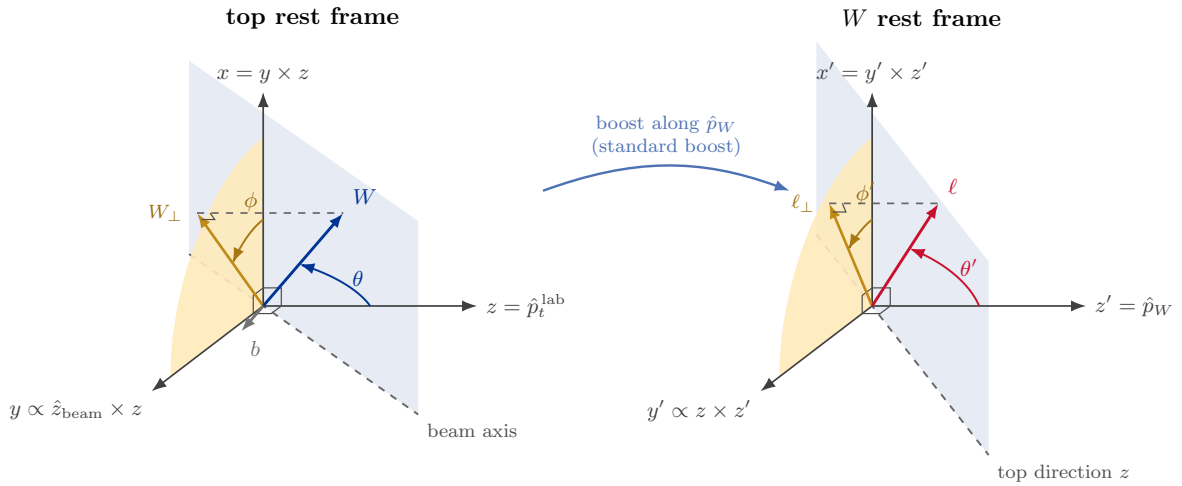



**Figure 2:** The $hh$ sample at parton, gen and reco level: $m_{b\bar{b}}$, the hadronic $W$ mass and the $\chi^2$-reconstructed top mass.

by building momenta from known angles and recovering them, which agrees to $10^{-14}$.




$x$–$z$ plane (contains the beam and $\hat{p}_t^{\text{lab}}$; right: the $x'$–$z'$ plane, contains $\hat{p}_W$ and the top direction) $xy$ plane, perpendicular to $z$

$\theta$: angle between $z$ and the $W$. $\phi$: **azimuth of the $W$ around $z$**: project the $W$ onto the $xy$ plane ($W_\perp$); $\phi$ is the angle from $x$ to $W_\perp$, toward $y$.
$(\theta', \phi')$: the same for the charged lepton in the $W$ rest frame, with $(x', y', z')$: $z' = \hat{p}_W$, $y' = z \times z'/|z \times z'|$ (in the $xy$ plane), $x' = y' \times z'$, i.e. $(x', y', z') = R(\phi, \theta, 0)\,(x, y, z)$.
The beam is not along $z$ ($z$ is the top direction); it lies in the $x$–$z$ plane with a negative $x$ component, since $y \propto \hat{z}_{\text{beam}} \times z$.
Hadronic top: same construction with the down-type quark in place of $\ell$.

**Figure 3:** Definition of the angles in the top rest frame (left) and in the $W$ rest frame (right); the tilde angles are defined in the same way for the hadronic top.

## 3.1 Validation

The $W$ helicity fractions obtained from the moments of $\cos\theta'$ (Table 5) reproduce the SM expectation at parton level, $F_0 = m_t^2/(m_t^2 + 2m_W^2) \approx 0.70$, $F_L \approx 0.30$, $F_R \approx 0$, on both the leptonic and the hadronic side. At gen and reco level they are distorted by acceptance and combinatorics; no unfolding is attempted. At parton level the leptonic and hadronic decay angles have the same distributions, as they must in the SM (Kolmogorov–Smirnov $p$-values 0.84, 0.27, 0.24, 0.54 for $\cos\theta$, $\phi$, $\cos\theta'$, $\phi'$). At reco level they differ even when every jet is correctly assigned ($\cos\theta'$ KS distance 0.097), because the two sides are measured differently: lepton cuts and the neutrino solution on one side, jet cuts and jet response on the other.

**Table 5:** $W$ helicity fractions from the moments of $\cos\theta'$.

| | $F_0$ | $F_L$ | $F_R$ |
|---|---|---|---|
| parton, $\ell$ | $0.715 \pm 0.013$ | $0.288 \pm 0.008$ | $-0.003 \pm 0.008$ |
| parton, down-type $q$ | $0.705 \pm 0.013$ | $0.298 \pm 0.008$ | $-0.003 \pm 0.008$ |
| gen, $\ell$ | $0.769 \pm 0.019$ | $0.107 \pm 0.012$ | $0.123 \pm 0.012$ |
| gen, down-type $q$ | $0.954 \pm 0.017$ | $0.078 \pm 0.011$ | $-0.032 \pm 0.011$ |
| reco, $\ell$ | $0.687 \pm 0.026$ | $0.134 \pm 0.017$ | $0.179 \pm 0.017$ |
| reco, down-type $q$ | $0.929 \pm 0.023$ | $0.086 \pm 0.015$ | $-0.014 \pm 0.015$ |

**Table 6:** Angle resolution with respect to parton level: correlation coefficient / median $|\Delta|$, for all reconstructed events and for events with all jets correctly assigned.

| angle | gen, all | gen, correct | reco, all | reco, correct |
|---|---|---|---|---|
| $Y$ | 0.90 / 0.16 | 0.99 / 0.05 | 0.89 / 0.18 | 0.98 / 0.06 |
| $\cos\Theta$ | 0.67 / 0.13 | 0.97 / 0.04 | 0.64 / 0.15 | 0.97 / 0.06 |
| $\cos\theta$ | 0.70 / 0.14 | 0.89 / 0.06 | 0.62 / 0.22 | 0.84 / 0.11 |
| $\phi$ | 0.48 / 0.36 | 0.82 / 0.12 | 0.40 / 0.52 | 0.89 / 0.16 |
| $\cos\theta'$ | 0.68 / 0.11 | 0.92 / 0.04 | 0.60 / 0.20 | 0.89 / 0.11 |
| $\phi'$ | 0.45 / 0.37 | 0.68 / 0.13 | 0.30 / 0.63 | 0.47 / 0.28 |
| $\cos\tilde{\theta}$ | 0.59 / 0.18 | 0.98 / 0.04 | 0.45 / 0.24 | 0.97 / 0.05 |
| $\tilde{\phi}$ | 0.40 / 0.46 | 0.88 / 0.09 | 0.34 / 0.69 | 0.90 / 0.09 |
| $\cos\tilde{\theta}'$ | 0.41 / 0.18 | 0.97 / 0.04 | 0.31 / 0.25 | 0.96 / 0.06 |
| $\tilde{\phi}'$ | 0.39 / 0.55 | 0.93 / 0.06 | 0.33 / 0.73 | 0.94 / 0.07 |

Reco angles track the parton angles when the jets are assigned correctly; the smearing of the inclusive distributions is dominated by combinatorics, not by the detector response.

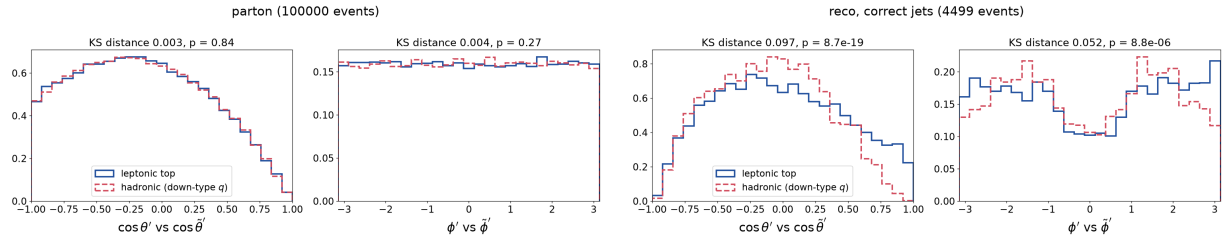



**Figure 4:** Leptonic versus hadronic $W$ decay angles at parton level (left) and at reco level with correctly assigned jets (right).

# 4 The density $f_{t\bar{t}}$ by OSDE

## 4.1 Basis and estimator

The basis is the paper's: for each top the 24 functions

$$M^{j_1 j_2}_{m'm}(\phi,\theta,\phi',\theta') = \frac{\sqrt{2j_1+1}\sqrt{2j_2+1}}{4\pi} D^{j_1}_{m'm}(\phi,\theta,0)\, D^{j_2}_{m0}(\phi',\theta',0), \qquad j_1 \le 1,\ j_2 \le 2, \tag{2}$$

times normalized Legendre polynomials $p_l(\cos\Theta) = \sqrt{(2l+1)/2}\,P_l$ with $l < 5$ and normalized Hermite *functions* $h_n(Y) = (2^n n!\sqrt{\pi})^{-1/2} H_n(Y) e^{-Y^2/2}$ with $n < 5$; the bare Hermite polynomials would be orthogonal only with the weight $e^{-Y^2}$. The product $B = M_{\mathbf{j}} \tilde{M}_{\tilde{\mathbf{j}}}\, p_l\, h_n$ has $24\times24\times5\times5 = 14400$ terms. The orthogonal-series estimate is $\hat{c} = \frac{1}{N}\sum_k B^*(\mathbf{x}_k)$ with its standard error, and $f_{t\bar{t}} = \sum \hat{c}\, B$, which is real because the set is closed under complex conjugation. The Wigner $d$ functions are computed in the Jacobi-polynomial form, which is stable to $j = 24$.

Three separate densities are fitted: $f^{\text{parton}}_{t\bar{t}}$ on parton-level $t\bar{t}$ angles and applied to parton-level $t\bar{t}$ and $hh$, and likewise $f^{\text{gen}}$ and $f^{\text{reco}}$ (plus $f^{\text{parton+trig}}$). Each is fitted on the fit sample and evaluated on the independent sample and on $hh$. The code was checked with unit tests (`tests/test_osde.py`, 14 tests): the Wigner $d^j_{m'm}$ agree with closed forms and with `sympy` to $10^{-12}$; the Gram matrices

of $M$, $p_l$ and $h_n$ equal the identity to $10^{-12}$; a toy density with known coefficients is recovered from $2 \times 10^5$ events with unit-width pulls; the fitted $f$ has an imaginary part below $10^{-10}$ of the real part; the leave-one-out formula below equals brute-force refits; and the extended basis of Section 6 reduces exactly to the paper's basis.

### 4.2 Self-inclusion bias and leave-one-out

Each fit event adds $\frac{1}{N}\sum_j |B_j(\mathbf{x}_k)|^2$ to its own value of $f$, because its own contribution to every coefficient is evaluated at its own position. With 14 400 terms this self term is not negligible (Table 7): the fit sample looked systematically more $t\bar{t}$-like than the independent sample until the self term was removed. Removing it exactly (leave-one-out, LOO) makes the fit sample agree with the independent sample in the fraction of negative $f$, in the mean surprisal and in the Kolmogorov–Smirnov distance; the difference was entirely this bias, and largest at reco level where $N$ is smallest. All plots use LOO values for the fit sample.

**Table 7:** Fit sample versus independent sample: the self-inclusion bias and its removal by leave-one-out (LOO).

| level | self term / $f$ | $f \leq 0$: fit / fit LOO / indep. | $\langle s \rangle$: fit / fit LOO / indep. | KS vs indep.: fit / LOO |
|---|---|---|---|---|
| parton | 7% | 0.6% / 1.5% / 1.6% | 11.98 / 12.08 / 12.07 | 0.041 / 0.010 |
| parton+trig | 8% | 0.7% / 2.0% / 1.6% | 11.81 / 11.91 / 11.90 | 0.048 / 0.008 |
| gen | 10% | 1.0% / 2.7% / 2.7% | 11.44 / 11.57 / 11.59 | 0.071 / 0.012 |
| reco | 16% | 1.2% / 4.5% / 4.5% | 11.33 / 11.53 / 11.53 | 0.103 / 0.012 |

### 4.3 Results

Table 8 summarizes the fits; Figures 5–10 show the one-dimensional projections of $f_{t\bar{t}}$ against the data and the distributions of $y = f_{t\bar{t}}$ for $t\bar{t}$ and $hh$. The negative values of $f$ are the Gibbs overshoot of a truncated series, as noted in the paper; they are rare for $t\bar{t}$ and more frequent for $hh$, which lives where $f_{t\bar{t}}$ is small. The surprisal $s' = -\ln(y - y_0)$ with $y_0 = 2.5 \min y$ makes every event finite, and the ROC curve is obtained by scanning a cut on $y$ (events with $f < y_{\text{cut}}$ are called $hh$-like). At parton level $f_{t\bar{t}}$ separates $hh$ from $t\bar{t}$ with AUC 0.653 (0.669 after the lepton trigger); at gen and reco level the separation almost vanishes (0.525 and 0.507). Understanding and curing this loss is the subject of Sections 10 and 11.

**Table 8:** Summary of the $f_{t\bar{t}}$ fits in the paper's basis. $t\bar{t}$ numbers are from the independent sample; AUC is the probability that an $hh$ event has lower $f_{t\bar{t}}$ than a $t\bar{t}$ event.

| level | $N_{\text{fit}}$ | $N_{\text{indep}}$ | $N_{hh}$ | $\lvert\hat{c}\rvert > 3\sigma$ | $\int f$ | $f \leq 0$: $t\bar{t}$ / $hh$ | $\langle s \rangle$: $t\bar{t}$ / $hh$ | AUC |
|---|---|---|---|---|---|---|---|---|
| parton | 100000 | 10000 | 20000 | 186 | 1.006 | 1.6% / 5.9% | 12.07 / 12.56 | 0.653 |
| parton+trig | 71322 | 7170 | 10149 | 255 | 1.000 | 1.6% / 6.8% | 11.90 / 12.45 | 0.669 |
| gen | 41317 | 4134 | 4416 | 330 | 0.985 | 2.7% / 4.3% | 11.59 / 11.66 | 0.525 |
| reco | 24030 | 2440 | 1878 | 253 | 0.987 | 4.5% / 4.8% | 11.53 / 11.56 | 0.507 |

### 4.4 The Hermite scale

The width of the Hermite functions is a free parameter of the fit; the paper uses $s = 1$ and so does this note by default. The $Y$ distribution is narrower at gen and reco level (std $\approx 0.8$), and $s = 1$

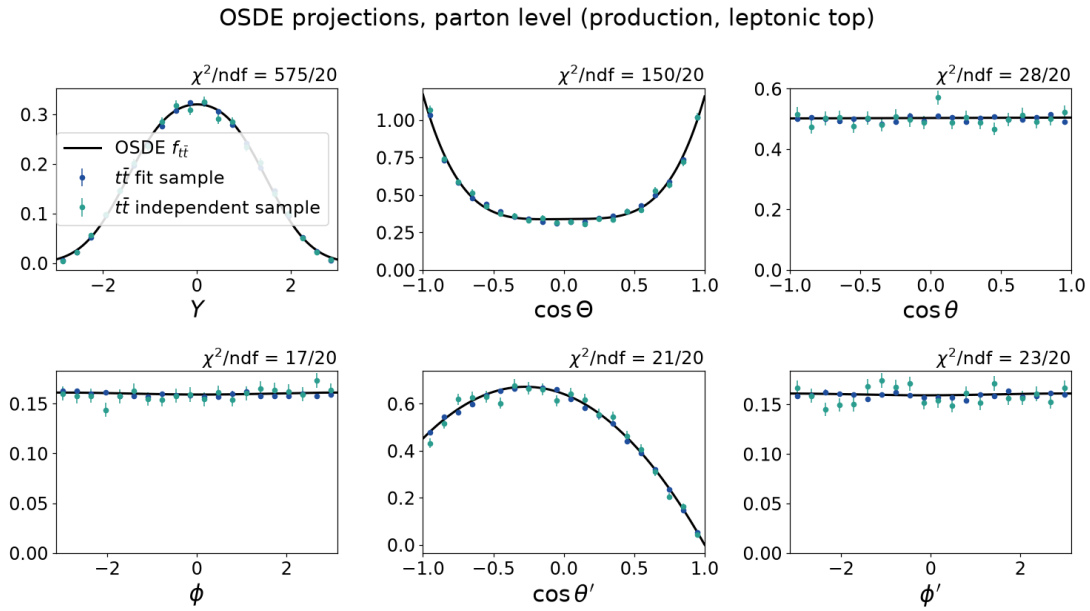



**Figure 5:** $f_{t\bar{t}}^{\text{parton}}$: one-dimensional projections (black) against the fit sample, production angles and leptonic top.

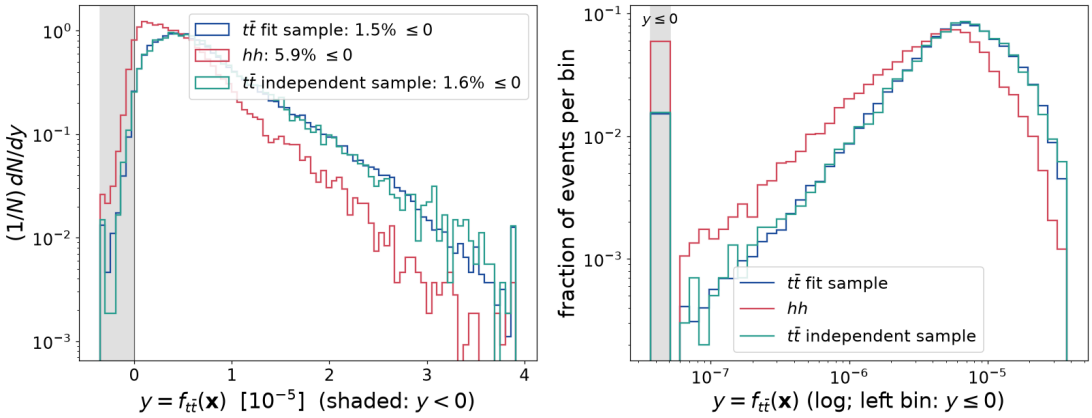



**Figure 6:** $y = f_{t\bar{t}}^{\text{parton}}$ for $t\bar{t}$ (independent sample and fit sample with LOO) and $hh$, linear (left, shaded: $y < 0$) and logarithmic (right; the left-most bin holds all events with $y \leq 0$).

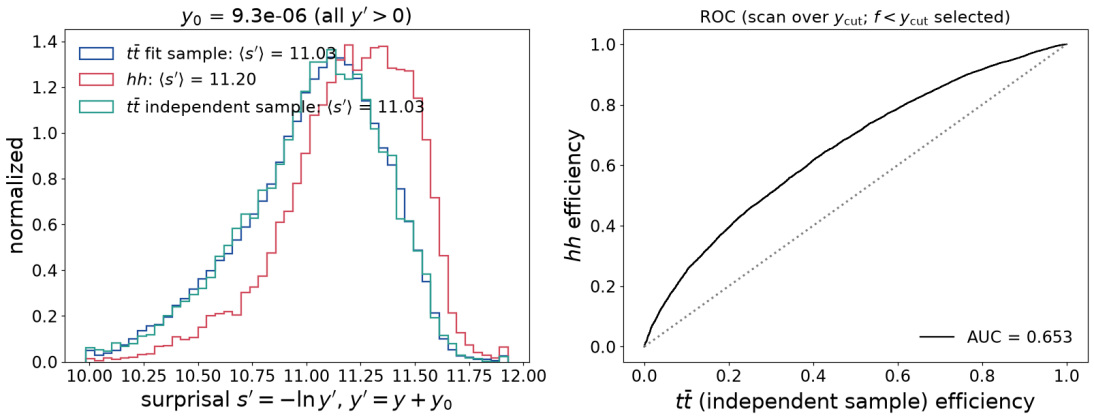



**Figure 7:** parton level: surprisal $s' = -\ln(y - y_0)$ (left) and the ROC curve from a scan over $y_{\text{cut}}$ (right).

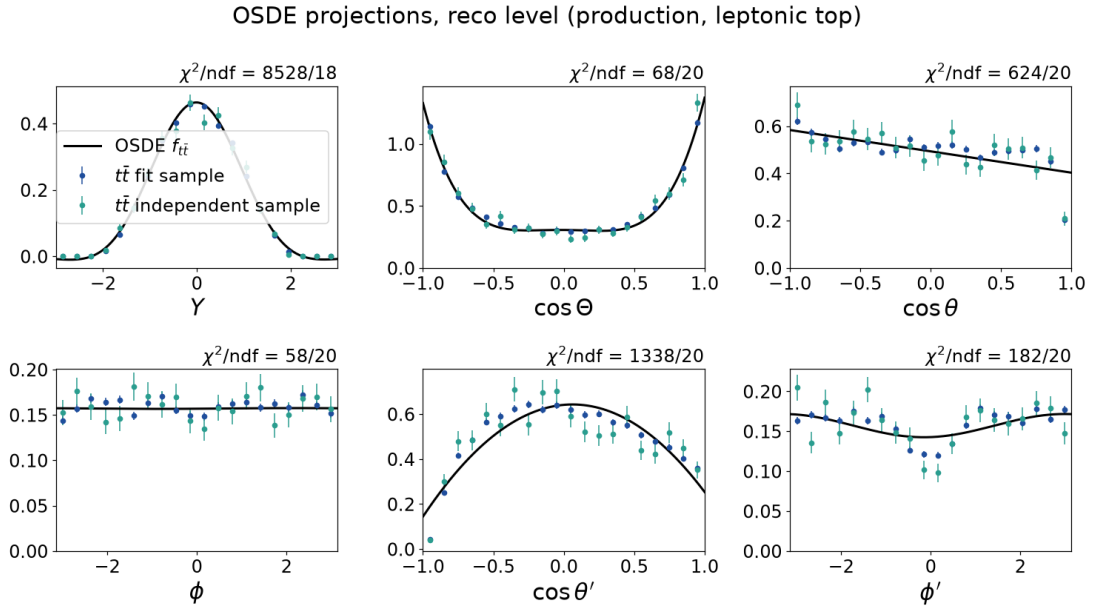



**Figure 8:** $f_{t\bar{t}}^{\text{reco}}$: one-dimensional projections (black) against the fit sample, production angles and leptonic top.

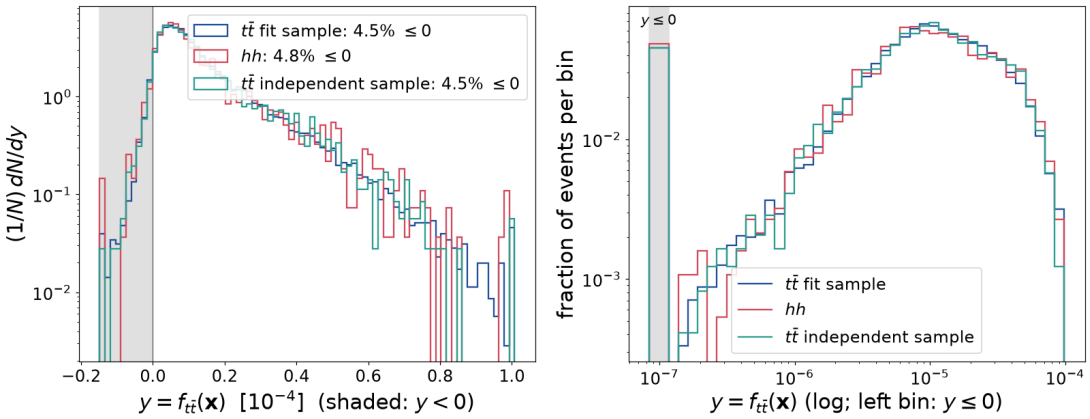



**Figure 9:** $y = f_{t\bar{t}}^{\text{reco}}$ for $t\bar{t}$ (independent sample and fit sample with LOO) and $hh$, linear (left, shaded: $y < 0$) and logarithmic (right; the left-most bin holds all events with $y \leq 0$).

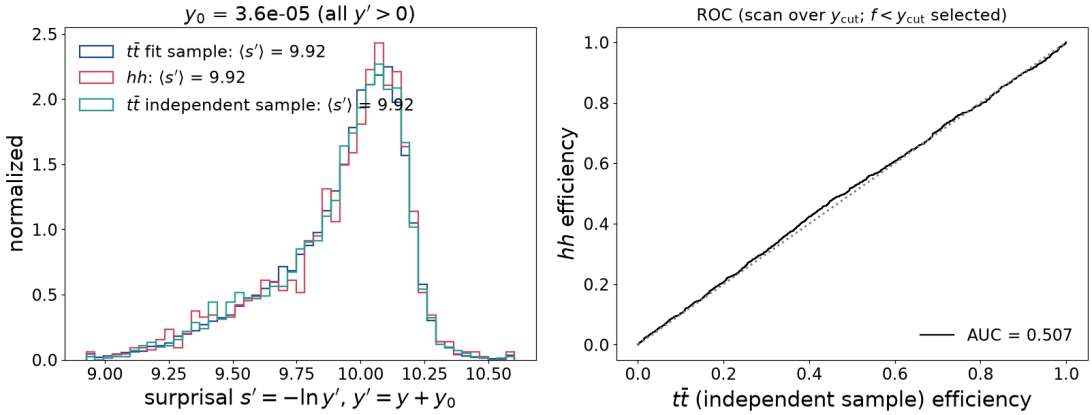



**Figure 10:** reco level: surprisal $s' = -\ln(y - y_0)$ (left) and the ROC curve from a scan over $y_{\text{cut}}$ (right).

then fits poorly: $f$ overshoots below zero at large $|Y|$ (Table 9). The extended basis of Section 6 uses $s = \mathrm{std}(Y)$.

**Table 9:** $\chi^2$/ndf of the $Y$ projection against the fit sample for different Hermite scales $s$.

| level | $s = 1.0$ | $s = 0.9$ | $s = 0.8$ | $s = 0.7$ | $s = \mathrm{std}(Y)$ |
|---|---|---|---|---|---|
| parton | 660/24 | 998/24 | 379/24 | 507/24 | 230/24 |
| parton+trig | 172/24 | 904/24 | 1013/24 | 117/24 | 111/24 |
| gen | 2298/20 | 2956/20 | 548/20 | 728/20 | 548/20 |
| reco | 4485/21 | 3864/21 | 721/21 | 361/21 | 515/21 |

## 4.5 The discriminator $t$

The paper's variable $t$ is a monotonic rescaling of the surprisal: $y' = y - y_0$, $s' = -\ln y'$ and $t = (s'_1 - s')/(s'_1 - s'_0) \in [0, 1]$, so that $t \to 0$ is less $t\bar{t}$-like. Two conventions had to be inferred from the paper's figure. First $y_0$: its axis starts at about $2.5 \times \min y$ and its $t$ distribution is populated from 0, which needs $y_0$ well below $\min y$; $y_0 = 2.5 \min y$ is used here. The shape of $t$ depends on this choice (Figure 12), the ROC curve and AUC do not, since the map is monotonic. Second, the endpoints $s'_0$ and $s'_1$ are taken as the 0.5% and 99.5% quantiles of $s'$ over $t\bar{t}$ and $hh$ together, with the events beyond put into the first and last bins, so that $t$ fills $[0, 1]$ with no empty bins. Figure 11 shows $y$ and $t$ in the style of the paper at parton level; the same figures for the other levels are in `osde/`.

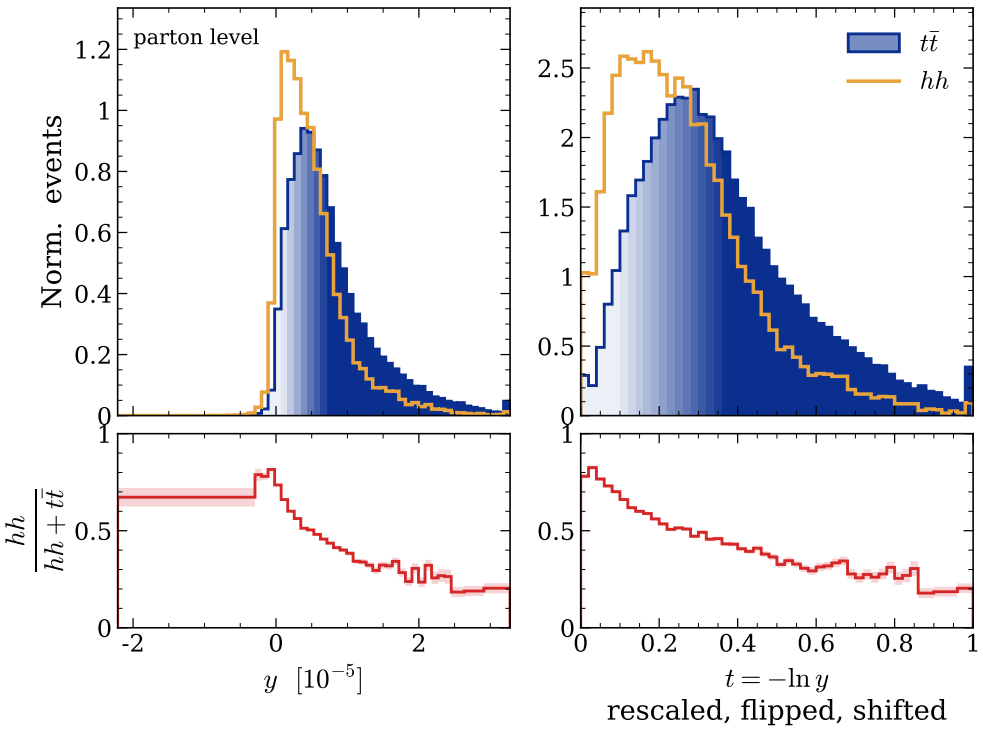



**Figure 11:** $y = f_{t\bar{t}}^{\mathrm{parton}}$ and $t$ for $t\bar{t}$ (fit sample, LOO) and $hh$ at unit area, with the ratio $hh/(hh + t\bar{t})$ below, drawn as in the paper.

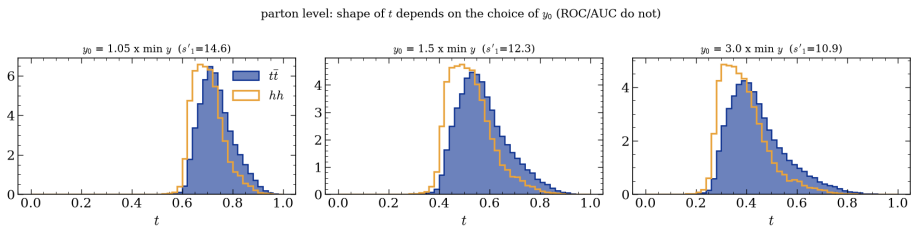



**Figure 12:** Shape of $t$ for different choices of $y_0$ at parton level.

# 5 How many coefficients matter?

## 5.1 Criterion

A term is kept if and only if its coefficient is significant, $|\hat{c}|/\sigma(\hat{c}) > t$, with the constant always kept. Conjugate partners have equal $|\hat{c}|$ and $\sigma$, so they are kept or dropped together and $f$ stays real (checked). The Kronmal–Tarter rule is $t = \sqrt{2}$: dropping a term costs $|c|^2$ in integrated squared error, keeping it costs its variance $\sigma^2$, so a term is kept if $|\hat{c}|^2 > 2\sigma^2$. With 14 400 mostly null terms this still admits about 16% of the pure-noise terms, so $t$ is instead chosen by five-fold cross-validation. For an orthonormal basis the cross-validated error needs no integrals: $\mathrm{ISE} - \int f^2 = \sum_{\mathrm{kept}} |\hat{c}|^2 - 2\,\mathrm{Re} \sum_{\mathrm{kept}} \hat{c}\,\hat{c}^*_{\text{held-out}}$. The minimum is at $t = 2.5$ and keeps 326, 385, 459, 342 of the 14 400 terms at parton, parton+trig, gen and reco level (Figure 13); the curve is flat out to $5\sigma$ (54, 108, 153, 127 terms).

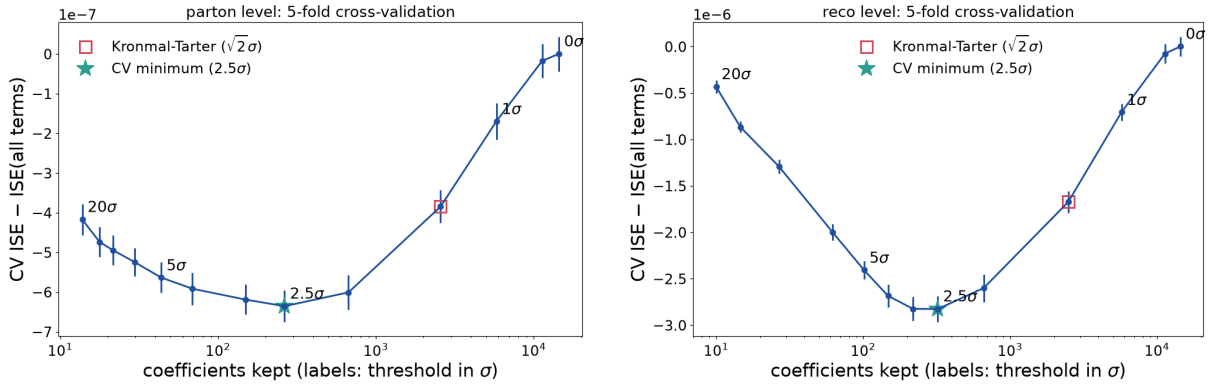



**Figure 13:** Cross-validated error versus the number of coefficients kept, parton (left) and reco level (right).

## 5.2 What the significant terms are

Table 11 lists the leading terms at parton level. They are the production angle ($P_2$, $P_4$ of $\cos\Theta$, i.e. the forward–backward peaking of $t$-channel $gg \to t\bar{t}$), the width of the rapidity distribution ($h_2$, $h_4$ of $Y$), and the $W$ helicity on each side ($j_1 = 0$, $j_2 = 1, 2$: functions of $\cos\theta'$ alone, i.e. the helicity fractions $F_0$, $F_L$, $F_R$), followed by their products. Table 12 classifies all terms above $5\sigma$ by their physics content: at parton level the $t\bar{t}$ spin correlations are resolved, with imaginary parts consistent with zero (maximum $1.4\sigma$) as expected without CP violation; at reco level the spin correlations are no longer resolved and acceptance-induced "polarization" terms appear instead.

**Table 10:** Performance of the truncated $f_{t\bar{t}}$. The simplified density has fewer negative values and the same or better $hh$ separation with 1–3% of the terms.

| level | terms kept | number | AUC | $f \leq 0$ ($t\bar{t}$ indep.) | $f \leq 0$ ($hh$) |
|---|---|---|---|---|---|
| parton | all (paper) | 14400 | 0.653 | 1.6% | 5.9% |
| | $\sqrt{2}\sigma$ (K–T) | 2629 | 0.658 | 1.0% | 4.8% |
| | CV best | 326 | 0.665 | 0.2% | 2.4% |
| | $5\sigma$ | 54 | 0.669 | 0.1% | 2.1% |
| parton+trig | all (paper) | 14400 | 0.669 | 1.6% | 6.8% |
| | $\sqrt{2}\sigma$ (K–T) | 2756 | 0.677 | 1.2% | 5.7% |
| | CV best | 385 | 0.685 | 0.4% | 3.4% |
| | $5\sigma$ | 108 | 0.691 | 0.4% | 3.7% |
| gen | all (paper) | 14400 | 0.525 | 2.7% | 4.3% |
| | $\sqrt{2}\sigma$ (K–T) | 2725 | 0.526 | 1.9% | 2.9% |
| | CV best | 459 | 0.525 | 0.9% | 1.3% |
| | $5\sigma$ | 153 | 0.527 | 1.0% | 1.5% |
| reco | all (paper) | 14400 | 0.507 | 4.5% | 4.8% |
| | $\sqrt{2}\sigma$ (K–T) | 2551 | 0.508 | 2.7% | 2.8% |
| | CV best | 342 | 0.505 | 1.1% | 1.5% |
| | $5\sigma$ | 127 | 0.507 | 1.5% | 1.9% |

**Table 11:** Leading terms at parton level, after the constant. $M$: leptonic top; $\tilde{M}$: hadronic top; $j_1 = 0$: $W$ helicity; $j_1 = 1$: top spin.

| # | term | $\lvert\hat{c}\rvert/\sigma$ | physics |
|---|---|---|---|
| 1 | $P_2(\cos\Theta)$ | 115 | production angle |
| 2 | $M^{01}_{0,0}$ | 88 | W helicity (lep) |
| 3 | $\tilde{M}^{01}_{0,0}$ | 87 | W helicity (had) |
| 4 | $M^{02}_{0,0}$ | 80 | W helicity (lep) |
| 5 | $\tilde{M}^{02}_{0,0}$ | 79 | W helicity (had) |
| 6 | $h_2(Y)$ | 59 | rapidity |
| 7 | $P_4(\cos\Theta)$ | 34 | production angle |
| 8 | $M^{01}_{0,0} \times P_2(\cos\Theta)$ | 32 | W helicity (lep); production angle |
| 9 | $\tilde{M}^{01}_{0,0} \times P_2(\cos\Theta)$ | 31 | W helicity (had); production angle |
| 10 | $M^{02}_{0,0} \times P_2(\cos\Theta)$ | 29 | W helicity (lep); production angle |

**Table 12:** Physics content of the terms above $5\sigma$.

| content of the term | parton | parton+trig | gen | reco |
|---|---|---|---|---|
| production angle + top pol. $\times$ $W$ hel. interference | – | 10 | 30 | 26 |
| rapidity + top pol. $\times$ $W$ hel. interference | – | 15 | 24 | 12 |
| top pol. $\times$ $W$ hel. interference + $t\bar{t}$ spin correlation | 16 | 12 | – | – |
| W helicity + production angle | 12 | 7 | 8 | 6 |
| production angle + rapidity + top polarization | – | 1 | 12 | 12 |
| production angle + rapidity + top pol. $\times$ $W$ hel. interference | – | 7 | 11 | 11 |
| W helicity + production angle + rapidity | 1 | 2 | 11 | 11 |
| W helicity + top pol. $\times$ $W$ hel. interference | – | 10 | 4 | 4 |
| **total** | **53** | **107** | **152** | **126** |

### 5.3 Why fewer terms do better: estimation noise

A density with about 20 terms separates $hh$ from $t\bar{t}$ *better* than the full 14 400-term expansion (0.670 against 0.653 at parton level). This is not a fluctuation but estimation noise. Each $\hat{c}_k$ is a sample mean over $N$ events and carries noise $\sigma_k \propto 1/\sqrt{N}$; most of the true coefficients are essentially zero, so those terms contribute only noise. Summed over thousands of terms the noise is a random wobble on $\hat{f}(\mathbf{x})$, unrelated to $t\bar{t}$ versus $hh$, which scrambles the ranking of events. Dropping the insignificant terms removes the wobble at the cost of a small bias: the usual bias–variance trade-off. Two predictions follow, and both hold (Table 13, Figure 14): the gain of the truncation shrinks as $N$ grows, like $1/\sqrt{N}$, while the truncated $f$ is converged already at 12.5k events; and two independent fits on disjoint halves of the data rank the events almost identically with the truncated $f$ but not with the full one.

**Table 13:** Truncation gain versus the size of the fit sample, at parton level; the test sample is always the independent $t\bar{t}$ sample. Gain errors are ±0.002–0.003 (paired bootstrap).

| fit events $N$ | AUC, all terms | AUC, $\lvert c\rvert/\sigma > 10$ | gain | score correlation of two disjoint fits: all / truncated |
|---|---|---|---|---|
| 12k | 0.602 | 0.661 (10 terms) | +0.058 | 0.46 / 0.989 |
| 25k | 0.627 | 0.666 (16 terms) | +0.039 | 0.65 / 0.998 |
| 50k | 0.643 | 0.669 (19 terms) | +0.026 | 0.79 / 1.000 |
| 100k | 0.653 | 0.670 (23 terms) | +0.018 | – |

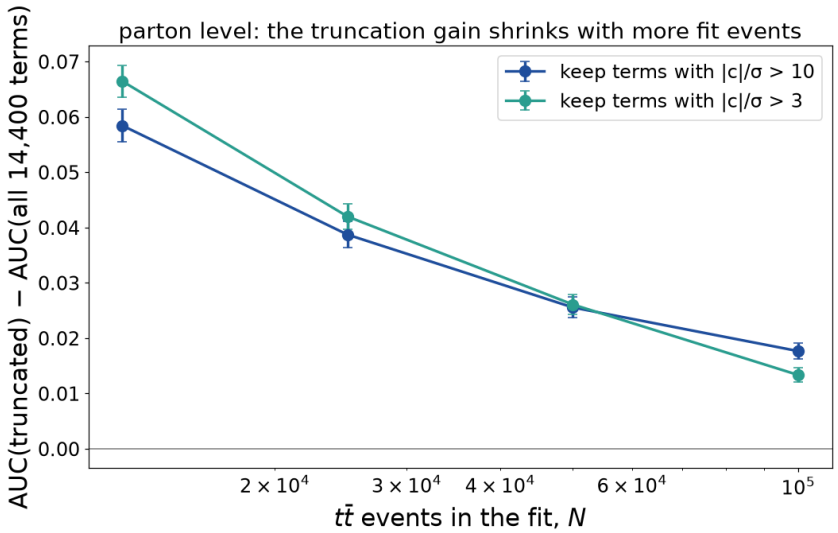



**Figure 14:** AUC gain of the truncated $f$ over the full expansion versus the number of fit events.

## 6 Higher orders for gen and reco: the extended basis

The paper's basis has $j_1 \leq 1$ and $j_2 \leq 2$: it is at most linear in $\cos\theta$, at most quadratic in $\cos\theta'$, and contains only the first harmonic in $\phi'$. This is complete for the physical decay distribution, and parton level confirms it, but the acceptance cuts create sharp edges ($\cos\theta \to 1$, $\cos\theta' \to -1$, $\phi' \approx 0$) that need higher orders. At gen level the one-dimensional projections of the paper-basis $f$ miss the data badly (Figure 15); a direct one-dimensional expansion at the same order misses it

in the same way (maximum difference 1.9%), while a higher-order expansion fits ($\chi^2$ for 25 bins: 1787, 1251, 572, 1048, 1000, 657 at the paper's order against 32, 13, 17, 15, 15, 14). So the order, not the fit, limits $f$.

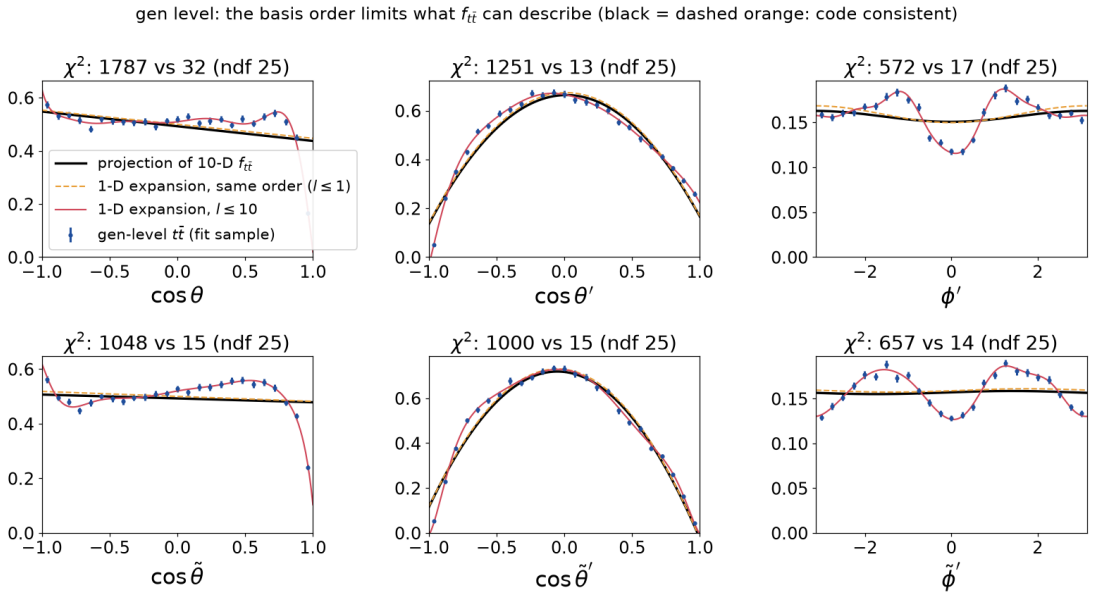



**Figure 15:** Gen level, paper basis: projections of $f$ (same order) against a higher-order one-dimensional expansion.

A one-dimensional cross-validated order scan per angle (Table 14) gives the orders each level needs. Parton level needs exactly the paper's orders apart from $\cos\Theta$ and $Y$; gen and reco need $j_{1,2}$ up to 13–24. A full product basis at those orders would have more than $10^6$ terms per top pair, so an *extended block basis* is used instead: the union of disjoint subsets of the full orthonormal product basis, which is still orthonormal so that the OSDE, the leave-one-out and the cross-validated truncation are unchanged. Block A is the paper's full product at the physics order; blocks L and H are high-order single-top blocks (all $(\theta, \phi)$ terms and all $m' = 0$ terms of one top with the other top constant); blocks P and Y are high-order $\cos\Theta$ and $Y$ terms, with the Hermite scale set to the standard deviation of $Y$. The orders are capped at 16; the resulting basis has 14436 (parton), 18936 (parton+trig), 32494 (gen), 49867 (reco) terms. The price is that correlations between the two tops stay at the physics order.

After the cross-validated truncation the extended basis describes every angle at every level with $\chi^2$/ndf between about 0.4 and 2 (Table 15, Figure 16), whereas the paper basis is good only at parton level. Yet this much better description of $t\bar{t}$ hardly changes the $hh$ separation at gen and reco level (Table 16): the limitation is not the density but the $t\bar{t}$-hypothesis reconstruction (Section 10). Without truncation the many high-order terms add noise, with up to 13% negative $f$.

## 7 Spherical-harmonic basis

As an alternative to the $M$ functions, products of spherical harmonics $Y_{l_1}^{m_1}(\theta, \phi)\, Y_{l_2}^{m_2}(\theta', \phi')$ with $l_1 \le 1$, $l_2 \le 2$ were used for each top: 36 functions instead of 24, 32 400 coefficients in all, with the same $p_l$ and $h_n$. The set is orthonormal and closed under conjugation (tested), and the same

**Table 14:** Order needed by each angle (one-dimensional cross-validated order: Legendre degree for cosines, harmonic for azimuths, Hermite order for $Y$) and the resulting requirement on $j_1$, $j_2$.

| angle | parton | parton+trig | gen | reco |
|---|---|---|---|---|
| $\cos\theta$ | 0 | 4 | 13 | 18 |
| $\phi$ | 0 | 2 | 4 | 4 |
| $\cos\theta'$ | 2 | 8 | 6 | 5 |
| $\phi'$ | 1 | 7 | 12 | 4 |
| $\cos\tilde{\theta}$ | 1 | 1 | 7 | 7 |
| $\tilde{\phi}$ | 0 | 0 | 4 | 3 |
| $\cos\tilde{\theta}'$ | 2 | 2 | 8 | 24 |
| $\tilde{\phi}'$ | 1 | 1 | 4 | 5 |
| $\cos\Theta$ | 10 | 6 | 6 | 13 |
| $Y$ | 10 | 10 | 8 | 24 |
| **needs** | $j_1 \geq 1,\ j_2 \geq 2$ | $j_1 \geq 4,\ j_2 \geq 8$ | $j_1 \geq 13,\ j_2 \geq 12$ | $j_1 \geq 18,\ j_2 \geq 24$ |

**Table 15:** $\chi^2$/ndf of the one-dimensional projections against the $t\bar{t}$ fit sample, paper and extended basis.

| level | basis | $Y$ | $\cos\Theta$ | $\tilde{\phi}'$ | $\cos\tilde{\theta}'$ | $\tilde{\phi}$ | $\cos\tilde{\theta}$ | $\phi'$ | $\cos\theta'$ | $\phi$ | $\cos\theta$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| parton | paper | 28.8 | 7.5 | 1.7 | 0.6 | 1.4 | 1.4 | 1.1 | 1.1 | 0.8 | 1.4 |
| | paper, CV | 28.6 | 6.9 | 1.9 | 0.6 | 1.4 | 1.6 | 1.3 | 1.1 | 0.9 | 1.4 |
| | extended | 0.5 | 0.5 | 1.5 | 0.4 | 1.1 | 1.2 | 0.9 | 0.8 | 0.6 | 1.2 |
| | extended, CV | 0.5 | 0.8 | 1.7 | 0.4 | 1.2 | 1.3 | 1.1 | 0.8 | 0.7 | 1.2 |
| parton+trig | paper | 7.9 | 2.7 | 1.3 | 0.3 | 1.1 | 1.1 | 47.0 | 2.5 | 0.8 | 2.7 |
| | paper, CV | 7.8 | 2.7 | 1.4 | 0.3 | 1.2 | 1.1 | 45.7 | 2.5 | 0.9 | 2.7 |
| | extended | 0.8 | 0.6 | 0.9 | 0.2 | 0.4 | 0.8 | 0.6 | 0.7 | 0.3 | 0.7 |
| | extended, CV | 0.8 | 1.0 | 1.4 | 0.3 | 1.2 | 1.1 | 1.8 | 1.4 | 0.4 | 0.9 |
| gen | paper | 1057.0 | 4.5 | 32.4 | 37.5 | 8.0 | 44.7 | 29.7 | 40.9 | 5.3 | 68.3 |
| | paper, CV | 906.9 | 4.6 | 32.5 | 39.4 | 8.2 | 44.7 | 30.1 | 40.9 | 5.6 | 68.3 |
| | extended | 16.1 | 1.3 | 0.4 | 0.5 | 0.2 | 0.2 | 0.2 | 0.4 | 0.1 | 0.7 |
| | extended, CV | 43.0 | 1.3 | 1.0 | 0.7 | 1.0 | 0.6 | 1.4 | 0.4 | 0.9 | 1.3 |
| reco | paper | 473.8 | 3.4 | 22.1 | 24.6 | 2.1 | 25.8 | 9.1 | 66.9 | 2.9 | 31.2 |
| | paper, CV | 473.3 | 3.6 | 22.1 | 24.2 | 2.2 | 26.6 | 9.1 | 68.8 | 2.9 | 31.2 |
| | extended | 9.6 | 0.4 | 0.3 | 0.7 | 0.2 | 1.1 | 0.5 | 0.4 | 0.1 | 0.6 |
| | extended, CV | 1.6 | 0.9 | 1.6 | 2.4 | 1.1 | 1.7 | 1.5 | 1.3 | 0.8 | 1.8 |

**Table 16:** Performance of the extended basis.

| level | terms | number | AUC ($hh$ vs $t\bar{t}$) | $f \leq 0$ ($t\bar{t}$ indep.) | $f \leq 0$ ($hh$) |
|---|---|---|---|---|---|
| parton | all terms | 14436 | 0.655 | 1.4% | 5.8% |
| | CV best | 333 | 0.666 | 0.3% | 2.5% |
| parton+trig | all terms | 18936 | 0.667 | 2.1% | 8.2% |
| | CV best | 498 | 0.684 | 0.4% | 4.3% |
| gen | all terms | 32494 | 0.528 | 5.7% | 8.4% |
| | CV best | 679 | 0.535 | 1.6% | 2.4% |
| reco | all terms | 49867 | 0.507 | 12.7% | 13.9% |
| | CV best | 323 | 0.512 | 1.4% | 1.8% |

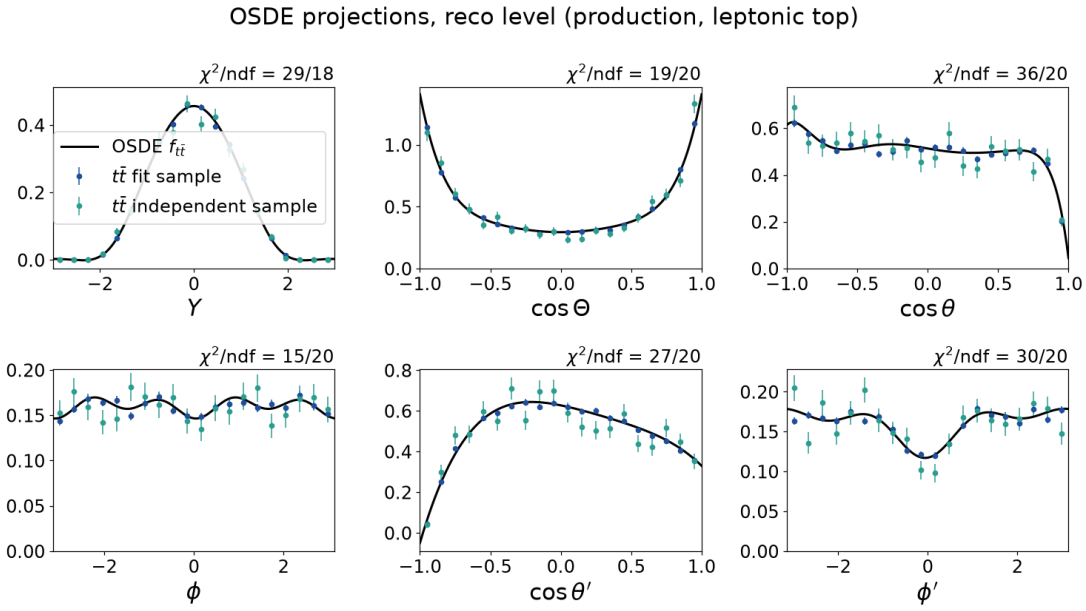



**Figure 16:** Extended basis at reco level after CV truncation: projections.

OSDE, leave-one-out and cross-validated truncation apply. The physics difference is that $M$ uses the spin-weighted $D^j_{m'm}(\phi,\theta,0)$: for example $d^1_{01} \propto \sin\theta$ is not a finite sum of $Y_{l0}$, so the SM decay density, which is finite in $M$, needs in principle infinitely many spherical-harmonic products. The figure of merit is the held-out integrated squared error on the independent $t\bar{t}$ sample, $\mathrm{ISE} - \int f^2 = \sum |\hat{c}|^2 - 2\langle \hat{f} \rangle_{\mathrm{indep}}$, lower being closer to the true density; differences between bases are computed on the same events (paired), which makes their errors small.

With all terms kept, the spherical harmonics are worse than $M$ at every level, by many standard deviations in $\Delta$ISE, simply because 32 400 terms are mostly noise. After the cross-validated truncation (Table 17, Figure 17) the two bases describe $t\bar{t}$ about equally well ($\Delta$ISE at parton level $+0.06 \pm 0.02$ in units of $10^{-6}$, i.e. $M$ is slightly better, about $3\sigma$; at gen and reco level they are equal within errors), and the spherical harmonics need fewer terms (150 against 326 at parton level). The extended basis is the best at gen and reco level. The $hh$ AUC hardly depends on the basis: once only the significant terms are kept, both bases describe the same few physical features.

## 8 Acceptance

To separate the effect of the selection from that of the reconstruction, Figure 18 shows the parton-level angles before any cut and after the gen and reco selections, so that the angle value of an event never changes and every shape difference is acceptance only; the angles measured at reco level are overlaid (dotted) to show what resolution and jet assignment add. The selection sculpts the angles in a few places. $\cos\theta' \to -1$ and $\phi' \approx 0$ are depleted because the lepton is emitted against the top's flight direction and fails the $p_T > 25$ GeV cut, already at gen level, so it is not an isolation effect. $|Y| \gtrsim 2$ is cut by the $|\eta| < 2.5$ acceptance. $\cos\theta$ and $\cos\tilde{\theta} \to +1$ are depleted because the $b$ quark then goes backward and is too soft for the jet cut. $\cos\Theta$ and the azimuthal angles have a nearly flat efficiency. The reco efficiency of $t\bar{t}$ after each cumulative cut is 1 lepton: 48.3%, $\geq$4 jets: 27.8%, $\geq$1 b-jet: 24.1%, $\chi^2$ reco: 24.0%; for $hh$ it is 27.0%, 10.7%, 9.4%, 9.4%. For $hh$ the

**Table 17:** Comparison of the bases on the independent $t\bar{t}$ sample. $\Delta$ISE$< 0$: closer to the true density than the paper's $M$ basis (paired). Worst $\chi^2$/ndf of the projections excludes $Y$.

| level | basis | terms | $\Delta$ISE vs $M$ $[10^{-6}]$ | worst $\chi^2$/ndf | $f \leq 0$ | AUC |
|---|---|---|---|---|---|---|
| parton | paper ($M$), all | 14400 | – | 7.5 | 1.6% | 0.653 |
| | paper ($M$), CV | 326 | – | 6.9 | 0.2% | 0.665 |
| | sph. harmonics, all | 32400 | $+1.05 \pm 0.07$ | 7.5 | 4.0% | 0.639 |
| | sph. harmonics, CV | 150 | $+0.06 \pm 0.02$ | 7.3 | 0.1% | 0.670 |
| | extended, all | 14436 | $-0.08 \pm 0.01$ | 1.5 | 1.4% | 0.655 |
| | extended, CV | 333 | $-0.02 \pm 0.01$ | 1.7 | 0.3% | 0.666 |
| parton+trig | paper ($M$), all | 14400 | – | 47.0 | 1.6% | 0.669 |
| | paper ($M$), CV | 385 | – | 45.7 | 0.4% | 0.685 |
| | sph. harmonics, all | 32400 | $+1.40 \pm 0.11$ | 8.2 | 4.5% | 0.648 |
| | sph. harmonics, CV | 269 | $+0.07 \pm 0.04$ | 7.4 | 0.3% | 0.688 |
| | extended, all | 18936 | $-0.04 \pm 0.04$ | 0.9 | 2.1% | 0.667 |
| | extended, CV | 498 | $-0.21 \pm 0.03$ | 1.8 | 0.4% | 0.684 |
| gen | paper ($M$), all | 14400 | – | 68.3 | 2.7% | 0.525 |
| | paper ($M$), CV | 459 | – | 68.3 | 0.9% | 0.525 |
| | sph. harmonics, all | 32400 | $+2.35 \pm 0.23$ | 68.3 | 6.6% | 0.520 |
| | sph. harmonics, CV | 555 | $-0.02 \pm 0.11$ | 68.3 | 1.3% | 0.525 |
| | extended, all | 32494 | $+0.85 \pm 0.15$ | 1.3 | 5.7% | 0.528 |
| | extended, CV | 679 | $-1.11 \pm 0.08$ | 1.4 | 1.6% | 0.535 |
| reco | paper ($M$), all | 14400 | – | 66.9 | 4.5% | 0.507 |
| | paper ($M$), CV | 342 | – | 68.8 | 1.1% | 0.505 |
| | sph. harmonics, all | 32400 | $+3.77 \pm 0.43$ | 66.9 | 9.9% | 0.507 |
| | sph. harmonics, CV | 279 | $-0.01 \pm 0.14$ | 64.5 | 1.5% | 0.510 |
| | extended, all | 49867 | $+6.34 \pm 0.32$ | 1.1 | 12.7% | 0.507 |
| | extended, CV | 323 | $-0.92 \pm 0.11$ | 2.4 | 1.4% | 0.512 |

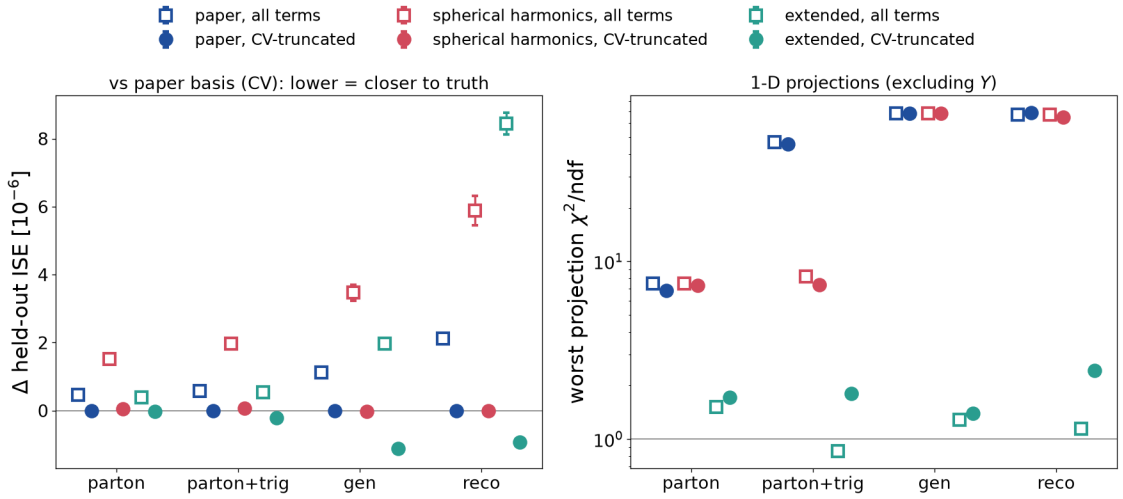



**Figure 17:** Paired held-out ISE difference with respect to the paper basis (left) and worst projection $\chi^2$/ndf (right), per level and basis.

dotted curves show the effect that dominates everything else: the $t\bar{t}$-hypothesis $\chi^2$ turns the flat parton-level $\cos\Theta$ into $t\bar{t}$-like peaks at $\pm 1$. The reconstruction, more than the acceptance, makes $hh$ look like $t\bar{t}$.

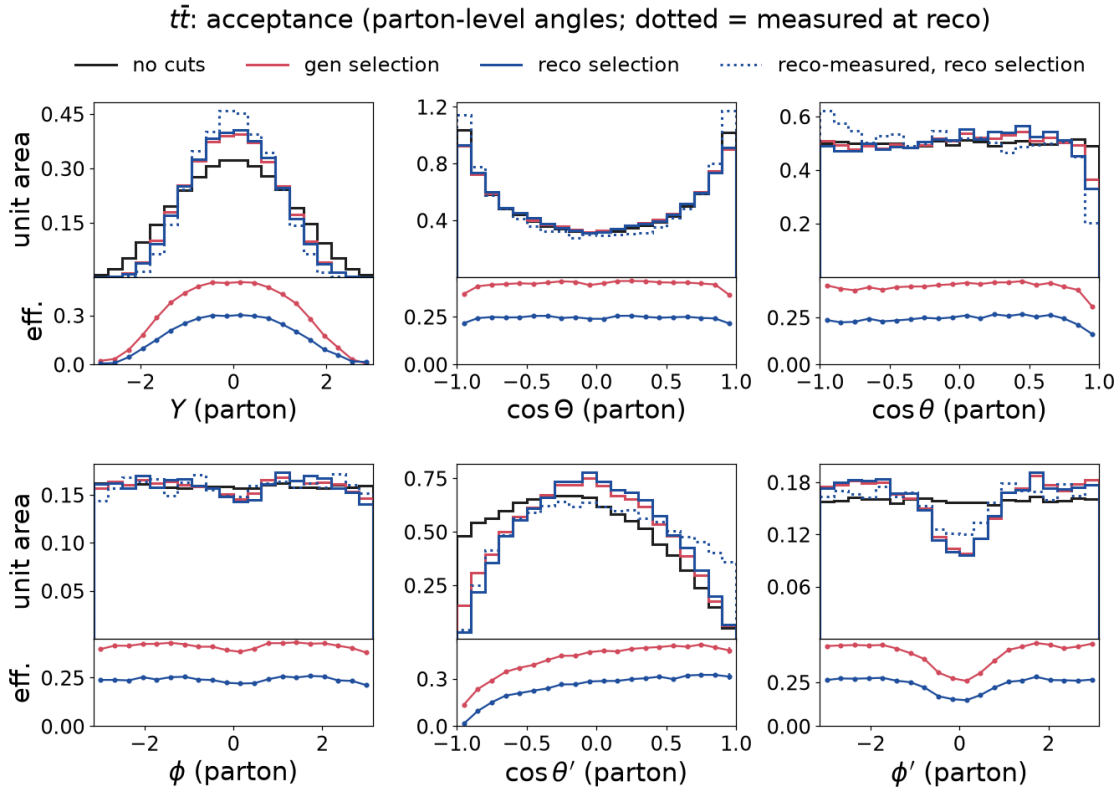



**Figure 18:** $t\bar{t}$: parton-level angles with no cuts, after the gen selection and after the reco selection (unit area), with the angles measured at reco level overlaid (dotted) and the efficiency below each panel.

# 9 Is the OSDE approach helping? Comparison with machine learning

## 9.1 Setup

Every method is evaluated on the same test events, the independent $t\bar{t}$ sample (10000 events) and a held-out 40% of $hh$ (8000 events); training uses the $t\bar{t}$ fit sample and 50% of $hh$, with 10% of each held out for validation. Errors are bootstrap errors; differences between methods are paired bootstraps on the same test events. The *unsupervised* methods are trained on $t\bar{t}$ only, exactly like $f_{t\bar{t}}$, on the same ten angles: a Gaussian-mixture density (32 components), an autoencoder scored by its reconstruction error, and a normalizing flow (RealNVP, 8 coupling layers). The *supervised* methods see labelled $t\bar{t}$ and $hh$ and therefore set an upper bound for the inputs: a boosted decision tree (BDT, scikit-learn histogram gradient boosting), a multi-layer perceptron (MLP) and an FT-Transformer on (a) the ten angles and (b) the angles plus the parton four-momenta, and a particle transformer with pairwise invariant-mass and $\Delta R$ attention biases on the momenta alone. To test whether $y$ helps a classifier, the same classifiers are trained with $y = f_{t\bar{t}}$ and $s'$ added to their inputs

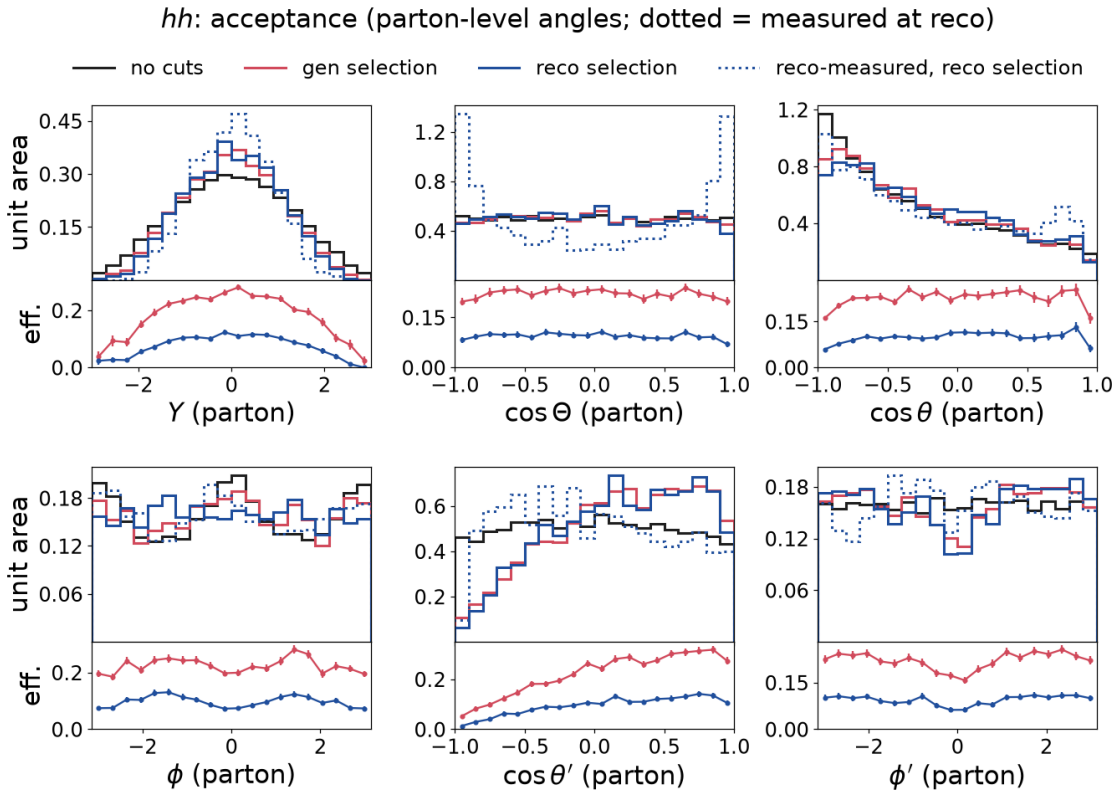



**Figure 19:** The same for $hh$.

(leave-one-out $y$ for the $t\bar{t}$ training events). Two controls must give AUC $\approx 0.5$: shuffled labels, and $t\bar{t}$ (fit sample) against $t\bar{t}$ (independent sample).

## 9.2 The methods in plain words

The unsupervised methods all do the same job as $f_{t\bar{t}}$: learn the $t\bar{t}$ density from $t\bar{t}$ events and flag events where it is low. The *Gaussian mixture* models the density as a sum of 32 Gaussian blobs in the ten-dimensional angle space, with positions and widths fitted to $t\bar{t}$. The *autoencoder* is a network that squeezes the ten angles through a few numbers and rebuilds them; trained on $t\bar{t}$, it rebuilds unusual events badly, but it is not a density. The *normalizing flow* learns a smooth, invertible map $z = T(\mathbf{x})$ that turns the $t\bar{t}$ distribution into a standard Gaussian; the density then follows from the change of variables, $f(\mathbf{x}) = \mathcal{N}(T(\mathbf{x}))\,|\det \partial T/\partial \mathbf{x}|$. The mass map of Section 11, $u = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{m-M}{W}$ with density $g_u\,du/dm$, is exactly a one-dimensional flow; RealNVP does the same in ten dimensions by stacking simple steps, each of which shifts and rescales half of the variables by amounts computed from the other half, so that the Jacobian stays cheap.

The supervised methods learn $P(hh\,|\,\mathbf{x})$ from labelled examples; the best any of them can do is the likelihood ratio $f_{hh}/f_{t\bar{t}}$, which $f_{t\bar{t}}$ alone can only approximate by acting as if $f_{hh}$ were flat. The *BDT* is a few hundred small decision trees, each correcting the mistakes of the previous ones. The *MLP* is a standard neural network of weighted sums with simple non-linearities in between. The *FT-Transformer* turns each input variable into a vector (a token) and lets every variable attend to every other, so it learns which combinations matter. The *particle transformer* uses the six particles' four-momenta as tokens and is told each pair's invariant mass and $\Delta R$, so it sees $m_{b\bar{b}} = m_h$ directly.

They win because they know what $hh$ looks like; the goal of $f_{t\bar{t}}$ is different, a $t\bar{t}$-only score that flags anything that is not $t\bar{t}$.

### 9.3 Results

**Table 18:** AUC ($hh$ vs $t\bar{t}$) of every method on the same test events, parton level and parton+trig level.

| inputs | method | AUC parton | AUC parton+trig |
|---|---|---|---|
| unsupervised | OSDE $f_{t\bar{t}}$, paper basis (all terms) | $0.652 \pm 0.004$ | $0.667 \pm 0.005$ |
| unsupervised | OSDE $f_{t\bar{t}}$, paper basis (CV-truncated) | $0.665 \pm 0.004$ | $0.683 \pm 0.005$ |
| unsupervised | Gaussian mixture density (32 comp.) | $0.583 \pm 0.005$ | $0.589 \pm 0.006$ |
| unsupervised | autoencoder (reconstruction error) | $0.528 \pm 0.004$ | $0.588 \pm 0.006$ |
| unsupervised | normalizing flow (RealNVP, 8 couplings) | $0.654 \pm 0.004$ | $0.675 \pm 0.005$ |
| angles | BDT | $0.881 \pm 0.002$ | $0.882 \pm 0.003$ |
| angles | MLP | $0.887 \pm 0.002$ | $0.896 \pm 0.003$ |
| angles | FT-Transformer | $0.884 \pm 0.002$ | $0.886 \pm 0.003$ |
| angles + y | BDT | $0.879 \pm 0.002$ | $0.882 \pm 0.003$ |
| angles + y | MLP | $0.886 \pm 0.002$ | $0.895 \pm 0.003$ |
| angles + y | FT-Transformer | $0.886 \pm 0.002$ | $0.885 \pm 0.003$ |
| angles + momenta | BDT | $1.000 \pm 0.000$ | $0.998 \pm 0.000$ |
| angles + momenta | MLP | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ |
| angles + momenta + y | BDT | $0.999 \pm 0.000$ | $0.998 \pm 0.000$ |
| angles + momenta + y | MLP | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ |
| momenta | Particle Transformer | $1.000 \pm 0.000$ | $1.000 \pm 0.000$ |
| control | control: BDT, shuffled labels | $0.491 \pm 0.005$ | $0.517 \pm 0.005$ |
| control | control: BDT, tt (fit) vs tt (independent) | $0.502 \pm 0.006$ | $0.503 \pm 0.007$ |

As an anomaly detector that has never seen $hh$, the CV-truncated $f_{t\bar{t}}$ is the best of the unsupervised methods, narrowly ahead of the normalizing flow (paired difference $+0.011 \pm 0.003$ at parton level and $+0.008 \pm 0.003$ at parton+trig level) and well ahead of the Gaussian mixture and the autoencoder. A supervised classifier on the same ten angles reaches 0.89, the upper bound for these inputs. Adding $y$ to the supervised classifiers changes their AUC by -0.002 to +0.002, within the uncertainty of $\pm 0.002$: $y$ carries no information that the classifier does not already learn from the angles. The momenta make the task trivial (AUC above 0.999), because they contain $m_{b\bar{b}} = m_h$ and the off-shell $W^*$ mass; this is the scale of what the angles alone leave out, and the starting point of Section 11. The controls give 0.491 and 0.502 at parton level. At parton+trig level the shuffled-label control is 0.517 $\pm 0.006$, about $3\sigma$ above 0.5; with some thirty AUCs computed, one such fluctuation is not surprising.

## 10 Why the separation washes out at gen level

### 10.1 Short answer

At parton level the $hh$ "tops" are not tops: each pseudo-top is the $W$ from one Higgs boson plus the $b$ quark from the other, with a mass of median 136 GeV, and one of the two $W$ bosons is far off shell (median 32 GeV); only 9% of the $hh$ events have both pseudo-tops within 30 GeV of $m_t$. This shows in the angles: $\cos\theta$ and $\cos\tilde{\theta}$ peak at $-1$ for $hh$ because the soft $W^*$ goes backward while the $b$ carries the pseudo-top's momentum, whereas they are flat for $t\bar{t}$; $\cos\Theta$ is isotropic for

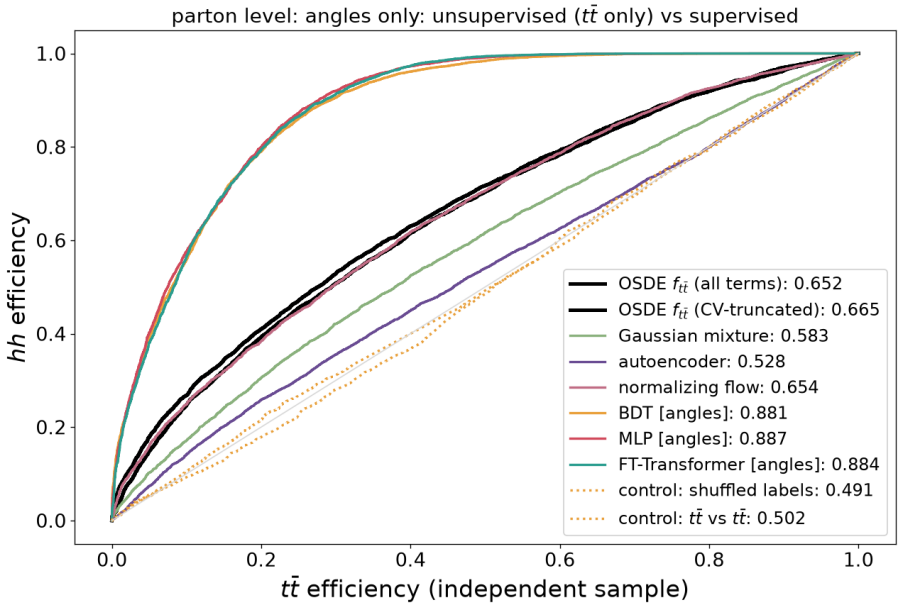



**Figure 20:** ROC curves at parton level, unsupervised and supervised methods on the ten angles.

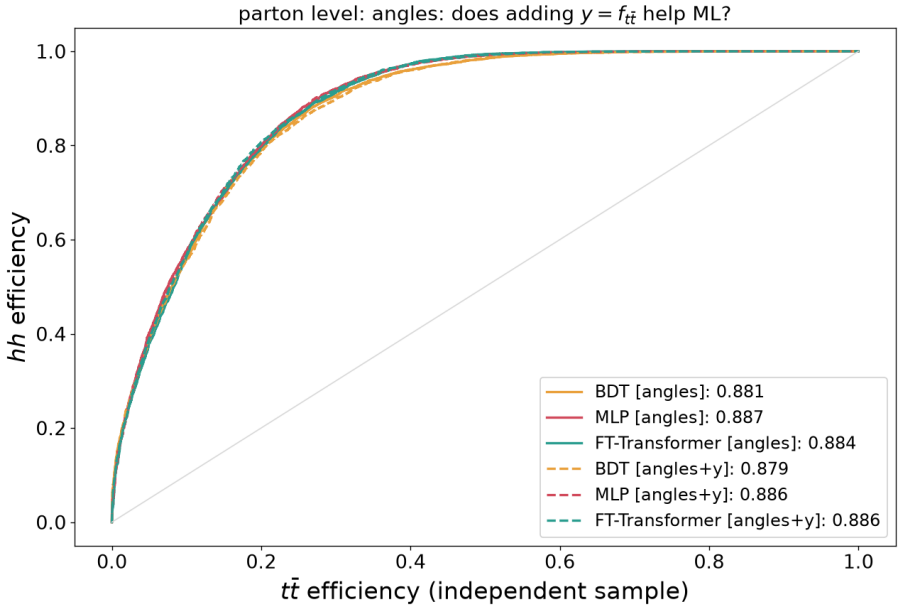



**Figure 21:** ROC curves at parton level: does adding $y$ help a supervised classifier?

$hh$ but forward–backward peaked for $t\bar{t}$ by $t$-channel top exchange; and $\cos\theta'$ has no $V-A$ zero at $+1$ for $hh$ (Figure 22). At gen level the $t\bar{t}$-hypothesis reconstruction removes these differences. The $\chi^2$ builds $m_W$ and $m_t$ from whatever jets fit best, picking the true four jets in only 8% of the $hh$ events against 34% of $t\bar{t}$, and the $m_W$-constrained neutrino puts the off-shell leptonic $W^*$ on shell (its $p_z$ is then off by a median 77 GeV, against 17 GeV for $t\bar{t}$). The information itself is still there at gen level: on the same events a BDT on observables that do not assume $t\bar{t}$ ($m_{b\bar{b}}$, $\Delta R_{bb}$, $E_T^{\text{miss}}$, $m_T$, ...) reaches AUC 0.87, with $m_{b\bar{b}}$ alone at about 0.76 (likelihood-ratio AUC); the $\chi^2$ value itself also differs (median 21.6 for $hh$ against 6.2 for $t\bar{t}$).

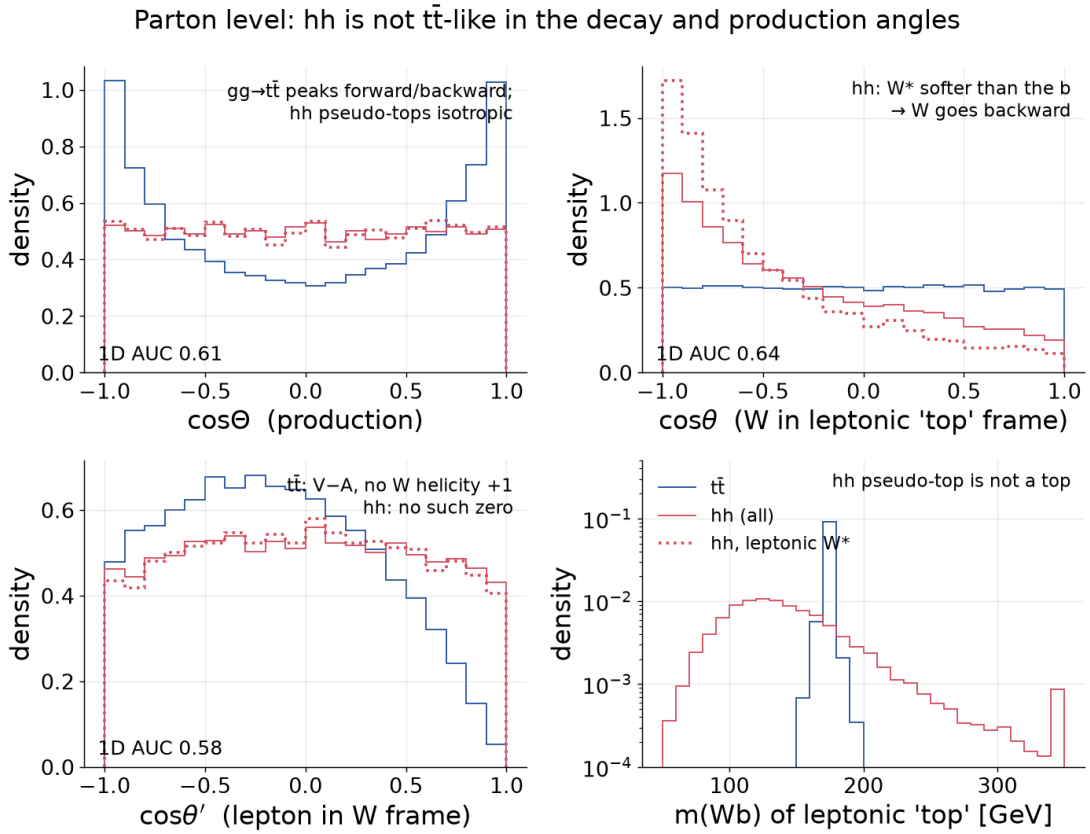



**Figure 22:** Parton level: what distinguishes $hh$ from $t\bar{t}$ in the angles and the pseudo-top masses.

## 10.2 Step-by-step decomposition

Two independent studies agree. The first follows the physics mechanism (Figures 22–26); the second measures the loss step by step from the parton level to the full gen-level reconstruction (Table 19, Figure 23). Each step re-fits $f_{t\bar{t}}$ on the $t\bar{t}$ events of that step, scored leave-one-out; the parton-level $f$ applied unchanged, and a supervised BDT on the ten angles, are shown alongside. An important technical point: the LHE partons are not the right reference at gen level, because Pythia's initial-state-radiation recoil kicks the whole hard system (the median $\Delta R$ between the LHE lepton and the final lepton is 0.16 for $t\bar{t}$ and 0.36 for $hh$). The "true" partons are therefore taken from the Pythia record, the decay products of the last-copy $t$, $W$ or $h$ after the initial-state radiation and before the final-state radiation.

**Table 19:** AUC along the chain from parton to gen level, all selected events ($hh = 1$). LOO: $f_{t\bar{t}}$ refit at that step and scored leave-one-out; fixed: the parton-level $f$ applied unchanged; BDT: supervised on the ten angles. Errors about $\pm 0.005$.

| step | $t\bar{t}$ / $hh$ | $f_{t\bar{t}}$ LOO | $f$ fixed | BDT |
|---|---|---|---|---|
| parton, all events | 100k / 20.0k | 0.650 | 0.643 | 0.881 |
| + gen selection | 41k / 4.4k | 0.659 | 0.690 | 0.866 |
| + ISR recoil (Pythia partons) | 41k / 4.4k | 0.652 | 0.691 | 0.785 |
| + FSR | 41k / 4.4k | 0.619 | 0.662 | 0.739 |
| $\chi^2$ jets + true $\nu$ | 41k / 4.4k | 0.556 | 0.600 | 0.659 |
| full gen $\chi^2$ ($\nu$ from MET) | 41k / 4.4k | 0.528 | 0.553 | 0.659 |

**Table 20:** The same on the subset of events whose four partons match distinct jets (39% of $t\bar{t}$, 12% of $hh$; errors about $\pm 0.013$).

| step | $f_{t\bar{t}}$ LOO | $f$ fixed | BDT |
|---|---|---|---|
| post-ISR parton | 0.599 | 0.664 | 0.743 |
| matched jets + gen $\ell$ + true $\nu$ | 0.608 | 0.659 | 0.753 |
| $\nu$ $p_z$ from $m_W$ (true $\nu$ $p_T$) | 0.552 | 0.575 | 0.753 |
| $\nu$ from MET + $m_W$ | 0.564 | 0.579 | 0.741 |
| $\chi^2$ jets + true $\nu$ | 0.605 | 0.655 | 0.706 |
| full $\chi^2$ | 0.554 | 0.562 | 0.698 |

The selection costs nothing, and the ISR recoil costs $f_{t\bar{t}}$ almost nothing (it costs the supervised BDT about 0.08, because the BDT's extra power at parton level relies on the exact $p_T = 0$ kinematics of the $t\bar{t}$ system). Final-state radiation costs 0.03. The largest loss, 0.06, is the choice of jets by the $\chi^2$: on the matched subset the jets and the $\chi^2$ cost nothing at all, so the whole loss is in the unmatched events, where the $\chi^2$ has to build the system from wrong jets; for $hh$ this is the soft $W^*$ quarks (median $p_T$ 27 GeV), which often make no jet. The neutrino costs 0.03 overall and 0.05 on matched events, almost all of it from the choice of the $p_z$ solution with the $m_t$ rule rather than from the missing-momentum resolution. Per angle (Figure 24), $\cos\Theta$ and $\cos\tilde{\theta}$ lose their separation at the jet choice, $\cos\theta'$ at the neutrino, and ISR recoil mainly costs $\phi$, $\tilde{\phi}$ and $Y$. At the end no single angle has an AUC above 0.57.

### 10.3 Conclusion

The angles are the right variables for $t\bar{t}$, but at gen and reco level they are computed under the $t\bar{t}$ hypothesis, which removes exactly what makes $hh$ different. On top of that, $f_{t\bar{t}}$ is unsupervised: it flags only events where $t\bar{t}$ is rare, and the fitted reconstruction moves $hh$ into the region where $t\bar{t}$ is dense. The information is not gone; it sits in the dimensions that the ten angles do not span. That is the subject of the next section.

## 11 Adding momentum to $y$: the resonance masses

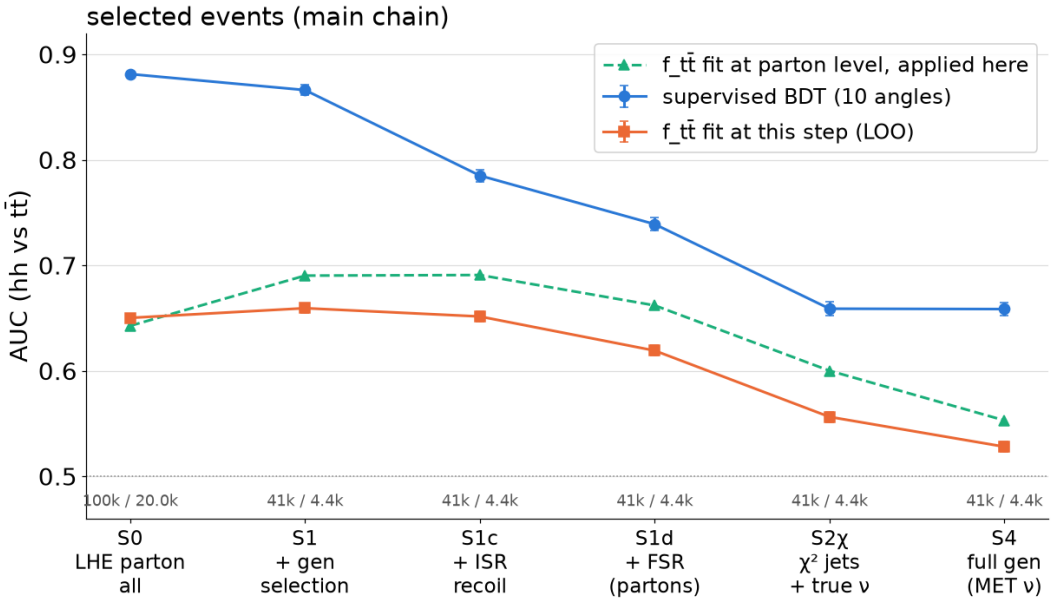



**Figure 23:** AUC along the chain from parton to gen level, all selected events.

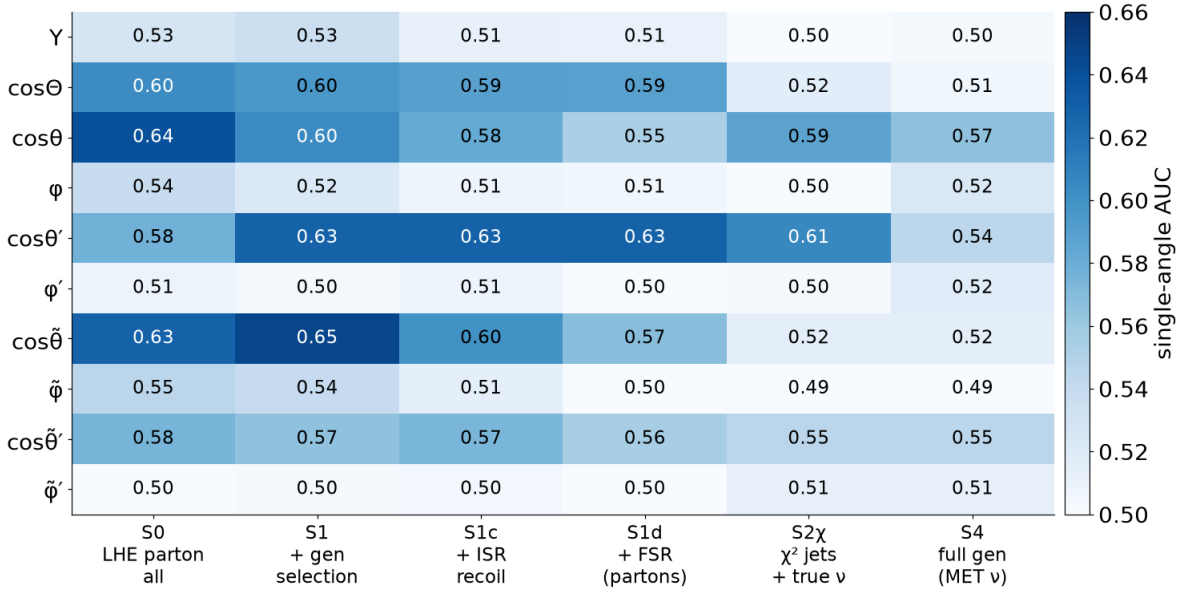



**Figure 24:** Single-angle AUC along the chain.

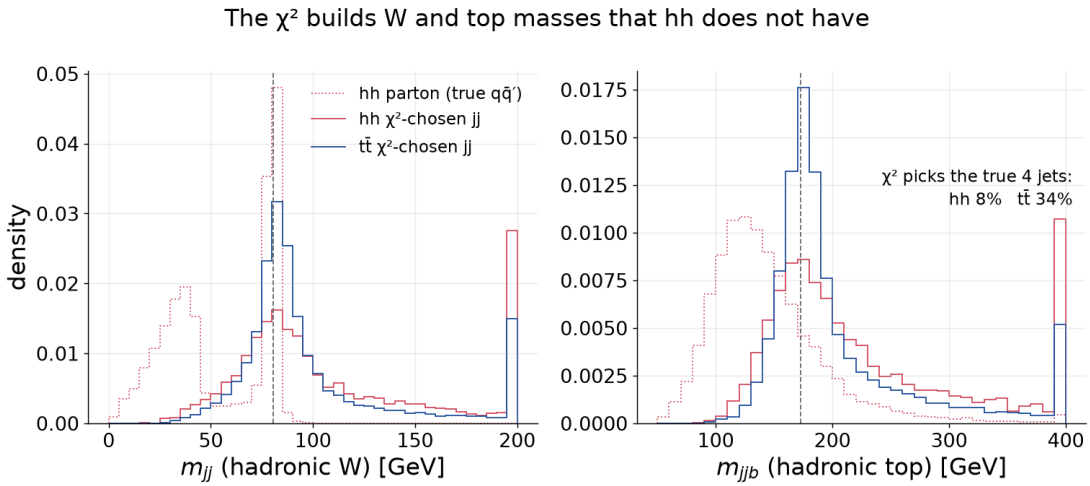



**Figure 25:** The $\chi^2$ builds $W$ and top masses that $hh$ does not have.

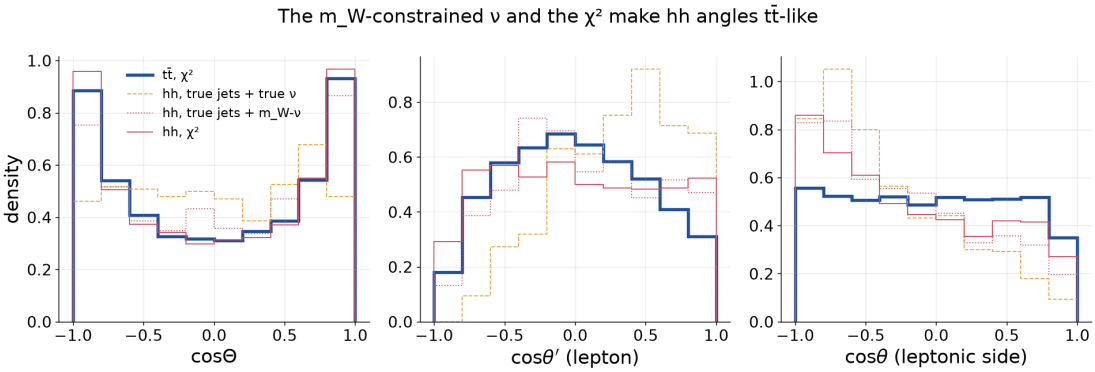



**Figure 26:** $hh$ angles become $t\bar{t}$-like: $\cos\Theta$, $\cos\theta'$ and $\cos\theta$ for $hh$ with true jets and the true neutrino, with the $m_W$-constrained neutrino, and after the full $\chi^2$, against $t\bar{t}$.

## 11.1 Motivation

The final state has six particles, i.e. 18 momentum components. They split exactly into the 10 paper variables ($Y$, $\Theta$ and the 8 decay angles), the 4 resonance masses $m_{W,\ell}$, $m_{W,h}$, $m_{t,\ell}$, $m_{t,h}$, and 4 production variables ($m_{t\bar{t}}$, the two components of the $t\bar{t}$ transverse momentum and an overall azimuth). For $t\bar{t}$ the four masses carry no information: each is a Breit–Wigner fixed by $M$ and $\Gamma$, and in the narrow-width limit the matrix element factorizes into the angular part times $\prod_k \mathrm{BW}(m_k)$, so the paper could rightly drop them. $hh$ is not on that shell: one $W$ is off shell and its pseudo-tops sit near 135 GeV (Figure 27). The difference between $hh$ and $t\bar{t}$ lives in exactly the dimensions that were dropped, which is the user's observation that the off-shell $W$ makes $hh$ a different final state from on-shell $t\bar{t}$.

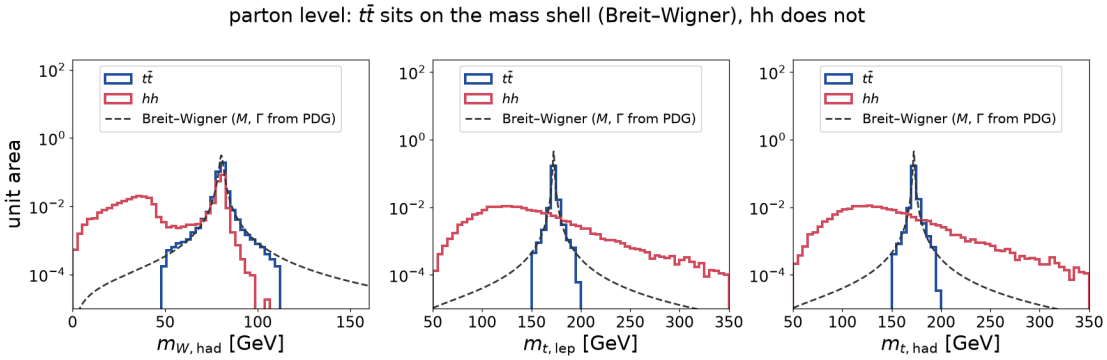



**Figure 27:** The dropped dimensions at parton level: $t\bar{t}$ (blue) sits on the Breit–Wigner (dashed; MadGraph cuts it at $\pm 15\Gamma$), $hh$ (red) is spread over 100 GeV.

## 11.2 The extension

The extension keeps $y(\Omega)$ as it is and multiplies it by one factor per resonance mass,

$$y_{\mathrm{ext}}(\Omega, m) = y(\Omega) \prod_{k=1}^{4} \mathrm{BW}_k(m_k), \tag{3}$$

the $t\bar{t}$ density on the full phase space. In the surprisal, from which $t$ is built, the factors simply add: $s_{\mathrm{ext}} = s(\Omega) + \sum_k \ln\left[(m_k^2 - M_k^2)^2 + M_k^2\Gamma_k^2\right] + \mathrm{const}$. At parton level $M$ and $\Gamma$ are the PDG values and nothing is fitted. At gen and reco level the detector resolution replaces $\Gamma$; the same peak shape is kept, with $M$ and the width $W$ taken as the median and half the interquartile range of $t\bar{t}$, i.e. two numbers per mass (Figure 28). A fitted correction to this shape, an orthonormal Legendre expansion in the Cauchy-mapped variable $u = \frac{1}{2} + \frac{1}{\pi}\arctan\frac{m-M}{W}$ with the Jacobian $du/dm$, was also tried (`study_salvage/salvage.py`): it changes the gen-level AUC by only +0.004, so it is not needed. At gen and reco level only three masses are available, because the neutrino solution fixes $m_{W,\ell} = m_W$ by construction.

## 11.3 Results

Table 21 adds the masses one at a time. At parton level the masses alone separate $hh$ completely, since $hh$ is never on the $t\bar{t}$ mass shell. At gen level the unsupervised score improves from 0.53 to 0.69,

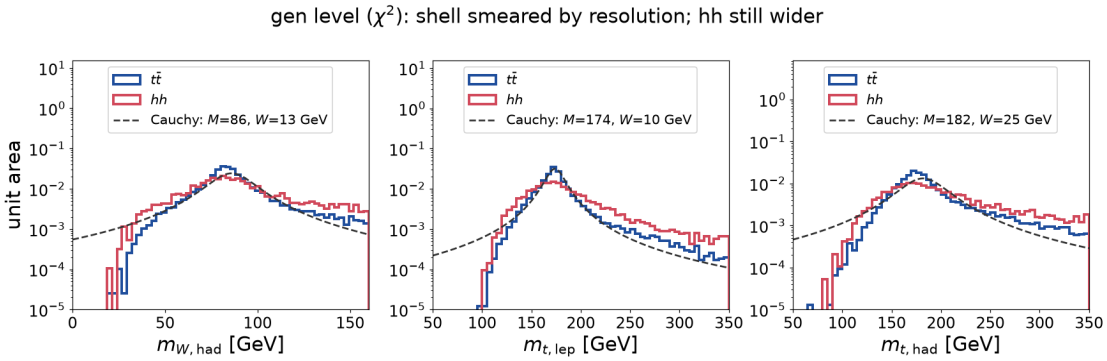



**Figure 28:** The same dimensions at gen level after the $\chi^2$ reconstruction: the $t\bar{t}$ shell is widened by the resolution (dashed: the two-number model); $hh$ is still wider, mainly in the tails.

which is more than the angles alone achieve at parton level (0.65): the parton-level performance is regained. Table 22 is the clean version of the same result at gen and reco level, with the $t\bar{t}$ test events from the independent sample and $M$, $W$ from 30k fit-sample events: reco level goes from 0.51 to 0.60. The gain is smaller at reco than at gen level because each mass separates less there (single-mass AUCs 0.574, 0.569, 0.575 against 0.628, 0.656, 0.637 at gen level), consistent with the $\chi^2$ having more jet combinations to fit at reco level and so building $t\bar{t}$-like masses more often for $hh$; the reco test sample is also small (2440 $t\bar{t}$ events, errors about $\pm 0.01$).

**Table 21:** AUC ($hh$ vs $t\bar{t}$), unsupervised, adding one Breit–Wigner factor at a time (test: half of 40k $t\bar{t}$ fit-sample events not used to set $M$, $W$, angles scored leave-one-out; 20k / 4.4k $hh$).

| | parton | gen ($\chi^2$) |
|---|---|---|
| angles only, $y(\Omega)$ (paper) | 0.648 | 0.528 |
| $+\, m_{W,\ell}$ | 0.786 | (fixed by the $\nu$ solution) |
| $+\, m_{W,h}$ | 0.993 | 0.631 |
| $+\, m_{t,\ell}$ | 0.999 | 0.683 |
| $+\, m_{t,h}$ (all masses) | 1.000 | 0.689 |

**Table 22:** The mass extension at gen and reco level on the independent $t\bar{t}$ sample. Widths $W$ (gen / reco): $m_{W,h}$ 13 / 12, $m_{t,\ell}$ 10 / 16, $m_{t,h}$ 25 / 25 GeV.

| level | $t\bar{t}$ / $hh$ | angles (all) | angles ($> 10\sigma$) | masses (BW) | angles ($> 10\sigma$) + BW |
|---|---|---|---|---|---|
| gen | 4134 / 4416 | 0.525 | 0.536 | 0.690 | 0.692 |
| reco | 2440 / 1878 | 0.507 | 0.518 | 0.600 | 0.602 |

## 11.4 What the angles add on top of the masses

In the product model the angles add nothing measurable for $hh$ once the mass factors are in (+0.002 at gen level, +0.002 at reco level). A supervised BDT shows how much is there in principle: trained on the angles and the masses against the masses alone, it gains $+0.029 \pm 0.004$ at gen level and $+0.037 \pm 0.015$ at reco level (paired bootstrap; `study_salvage/angles_vs_bw.py`). So a little angular information survives that the product form, which assumes angles and masses independent for $t\bar{t}$, does not use; a joint fit of angles and masses could recover part of it.

## 11.5 How many angular terms are needed once the masses are in?

With the mass factors in, the $hh$ separation is the same whatever the number of angular terms kept, even with none at all (Table 23): for $hh$ the masses do all the work. Even for the angles alone, about 20 terms beat all 14 400 (Section 5.3). The recommendation is therefore $y = [\text{few-term angular } f] \times \prod_k \mathrm{BW}(m_k)$. The small angular part is worth keeping: it costs almost nothing, and it is what would catch an on-shell signal with a different spin structure, which the masses cannot see (Section 12).

**Table 23:** AUC with the angular density truncated by coefficient significance, with and without the mass factors.

| angular terms kept | parton: terms | angles | +BW | gen: terms | angles | +BW |
|---|---|---|---|---|---|---|
| none (flat in the angles) | 0 | – | 1.000 | 0 | – | 0.689 |
| $\lvert c\rvert/\sigma_c > 40$ | 7 | 0.653 | 1.000 | 5 | 0.546 | 0.692 |
| $\lvert c\rvert/\sigma_c > 20$ | 16 | 0.662 | 1.000 | 14 | 0.555 | 0.692 |
| $\lvert c\rvert/\sigma_c > 10$ | 23 | 0.665 | 1.000 | 54 | 0.534 | 0.689 |
| $\lvert c\rvert/\sigma_c > 5$ | 54 | 0.663 | 1.000 | 153 | 0.528 | 0.688 |
| $\lvert c\rvert/\sigma_c > 3$ | 186 | 0.661 | 1.000 | 330 | 0.528 | 0.688 |
| CV (paper method) | 326 | 0.661 | 1.000 | 459 | 0.529 | 0.688 |
| all | 14400 | 0.648 | 1.000 | 14400 | 0.528 | 0.689 |

## 11.6 Realistic quark tagging

The $b$-tagging at reco level is the Delphes ATLAS parametrization and the $\chi^2$ assigns the $b$-jets, so that part is already realistic. What the paper assumes beyond it is knowing which $W$ jet is the down-type quark, and so far the jets were matched to the parton-level quarks, i.e. perfect up/down tagging. The only real handle is $c$-tagging: in $W \to cs$, 50% of the events at parton level, a $c$-tagged jet is up-type, so the other jet is the down-type $s$. Instead of a hard choice, a weight $w = P(\text{softer jet is down-type} \,|\, c\text{-tags})$ is computed by Bayes' rule from the $W \to cs$ and $W \to ud$ hypotheses, the tag efficiencies, and the prior that the softer jet is the down-type one (58% at parton level, 53% at reco level); then $y = w\, f(\Omega_{\text{down=softer}}) + (1-w)\, f(\Omega_{\text{down=harder}})$, the $t\bar{t}$ density of the event as observed, with $f$ unchanged. Tags are emulated from the true jet flavour (the Delphes jet flavour at reco level, the quark identity at parton level) at illustrative ATLAS-like working points, $\epsilon_c$ / $b$ mistag / light mistag of 30 / 11 / 1.4% and 45 / 30 / 8%, plus an ideal tagger.

Table 24 shows that realistic tagging costs at most 0.02 on the angles alone at parton level, nothing at reco level, and nothing at all once the masses are in, because the masses do not depend on which jet is down-type. The hadronic primed angles carry little of the $hh$ separation anyway.

**Table 24:** Effect of the down-type assignment (AUC $hh$ vs $t\bar{t}$, angular $f$ with the terms above $10\sigma$).

| down-type assignment | label right ($t\bar{t}$) | parton: angles | +BW | reco: angles | +BW |
|---|---|---|---|---|---|
| perfect up/down tag (paper) | 100% | 0.670 | 1.000 | 0.518 | 0.602 |
| ideal $c$-tag | 79% | 0.661 | 1.000 | 0.517 | 0.602 |
| $c$-tag loose (45 / 30 / 8%) | 65% | 0.653 | 1.000 | 0.517 | 0.602 |
| $c$-tag tight (30 / 11 / 1.4%) | 64% | 0.652 | 1.000 | 0.517 | 0.602 |
| no tag: average both labels | – | 0.650 | 1.000 | 0.517 | 0.602 |
| no tag: softer jet = down | 58% | 0.643 | 1.000 | 0.515 | 0.601 |

## 12 A BSM benchmark: $H \to t\bar{t}$

A supervised classifier trained on $hh$ learns what $hh$ looks like; $f_{t\bar{t}}$ learns only what $t\bar{t}$ looks like. The case for the unsupervised approach is a signal that nobody trained for. The benchmark is a heavy CP-even scalar, $gg \to H \to t\bar{t} \to b\bar{b}\,\ell\nu\,q\bar{q}'$ with $m_H = 500$ GeV and $\Gamma_H = 25$ GeV ($\Gamma/m = 5\%$, typical of a heavy two-Higgs-doublet scalar), generated with the `heft` $ggH$ coupling and SM-like decays (signal only, without the interference with QCD $gg \to t\bar{t}$). Unlike $hh$ it has on-shell tops and $W$ bosons: it differs from QCD $t\bar{t}$ only in the spin structure (scalar: isotropic $\cos\Theta$, like-helicity tops), in the rapidity and in $m_{t\bar{t}}$, not in the resonance masses. MadGraph's automatic width for this configuration came out absurd ($4 \times 10^5$ GeV), so the width is set by hand.

Table 25 compares, on the same events, the unsupervised scores with a supervised BDT trained on $hh$ versus $t\bar{t}$ and applied to both signals, and with a BDT trained on $H \to t\bar{t}$ itself as the ceiling of a dedicated search. The $m_{t\bar{t}}$ factor is a one-dimensional histogram density of the $t\bar{t}$ fit sample (median $m_{t\bar{t}}$: $t\bar{t}$ 472, $hh$ 307, $H \to t\bar{t}$ 502 GeV at parton level). Test events: 10000 $t\bar{t}$, 10000 $hh$ and 10000 $H \to t\bar{t}$ at parton level; 2440, 939 and 2951 at reco level.

**Table 25:** AUC of each score for $hh$ vs $t\bar{t}$ and for $H \to t\bar{t}$ vs $t\bar{t}$, parton and reco level (bootstrap errors).

| score | parton: $hh$ | parton: $H \to t\bar{t}$ | reco: $hh$ | reco: $H \to t\bar{t}$ |
|---|---|---|---|---|
| unsupervised: $f_{t\bar{t}}$ angles ($> 10\sigma$) | $0.672 \pm 0.003$ | $0.554 \pm 0.004$ | $0.523 \pm 0.012$ | $0.565 \pm 0.007$ |
| unsupervised: Breit–Wigner masses | $1.000 \pm 0.000$ | $0.506 \pm 0.004$ | $0.593 \pm 0.010$ | $0.536 \pm 0.008$ |
| unsupervised: $m_{t\bar{t}}$ density | $0.810 \pm 0.003$ | $0.557 \pm 0.004$ | $0.502 \pm 0.012$ | $0.487 \pm 0.007$ |
| unsupervised: angles + BW | $1.000 \pm 0.000$ | $0.512 \pm 0.004$ | $0.596 \pm 0.011$ | $0.550 \pm 0.008$ |
| unsupervised: angles + BW + $m_{t\bar{t}}$ | $1.000 \pm 0.000$ | $0.492 \pm 0.004$ | $0.584 \pm 0.011$ | $0.531 \pm 0.008$ |
| supervised, trained on $hh$: angles | $0.876 \pm 0.002$ | $0.523 \pm 0.004$ | $0.586 \pm 0.011$ | $0.505 \pm 0.008$ |
| supervised, trained on $hh$: angles + masses + $m_{t\bar{t}}$ | $1.000 \pm 0.000$ | $0.429 \pm 0.004$ | $0.687 \pm 0.011$ | $0.482 \pm 0.007$ |
| supervised, trained on $H \to t\bar{t}$: angles | $0.578 \pm 0.004$ | $0.611 \pm 0.004$ | $0.503 \pm 0.012$ | $0.553 \pm 0.007$ |
| supervised, trained on $H \to t\bar{t}$: angles + masses + $m_{t\bar{t}}$ | $0.212 \pm 0.003$ | $0.893 \pm 0.002$ | $0.460 \pm 0.011$ | $0.642 \pm 0.008$ |

The $hh$-trained classifier on the angles gives 0.52 on $H \to t\bar{t}$ at parton level, and with the masses and $m_{t\bar{t}}$ added 0.43; the unsupervised angles alone give 0.55 (0.57 at reco level), and with the mass factors and the $m_{t\bar{t}}$ density 0.49; a dedicated $H \to t\bar{t}$ classifier reaches 0.89.

Four things follow. First, a classifier trained on one signal can be worse than random on another: the $hh$-trained BDT learned that on-shell masses and high $m_{t\bar{t}}$ mean $t\bar{t}$, and $H \to t\bar{t}$ has both. Second, the unsupervised angular $f_{t\bar{t}}$ is weakly positive and model-agnostic; it sees the isotropic $\cos\Theta$ and the different spin structure, at parton and at reco level. Third, the mass factors add nothing for an on-shell signal and *dilute* the angular score, because the summed surprisal is dominated by the dimensions with the largest spread; for an on-shell signal the angular part alone, or a weighting of the factors, is the right score. Fourth, there is a limit to any density-based anomaly score: the $m_{t\bar{t}}$ resonance at 500 GeV sits on the bulk of the QCD $t\bar{t}$ distribution (median 472 GeV), so "low $t\bar{t}$ density" cannot flag it, whereas a dedicated supervised search can. Anomaly detection finds signals where the background is rare, not excesses where it is common. So there are signals for which $f_{t\bar{t}}$ beats the $hh$-trained classifier, but the margin is small; the strength of the approach is that it never does worse than random.

## 13 Pileup

All results above are for zero pileup. The reco-level study is repeated at mean pileup $\langle\mu\rangle = 50$ and 200 with the Delphes ATLAS pileup card (pileup-subtracted jets, the same object and selection

cuts), with the reco-level $f_{t\bar{t}}$ refitted at each pileup level and the mass extension applied. Gen level is unaffected by pileup by construction.

*Status: the pileup samples are being produced; this section is filled in automatically from* `pileup/results.json` *when the study completes.*

## 14 Open questions and next steps

- **Conventions to confirm with the paper.** $y_0 = 2.5\,\min y$ was inferred from the paper's figure; the sign convention in the paper's text and in its caption disagree.
- **Neutrino.** Solving the neutrino with the $m_{t,\ell}$ constraint instead of $m_W$ would free $m_{W,\ell}$, which is the off-shell $W^*$ in half of the $hh$ events, and would remove the largest single loss on correctly reconstructed events.
- **No single hypothesis.** Instead of committing to one $\chi^2$ assignment, evaluate $y = \sum_a w_a\, f_{t\bar{t}}(\Omega_a, m_a)$ summed over jet assignments and neutrino solutions, the density of the observed final state under $t\bar{t}$ in the spirit of the matrix-element method; this removes the $\chi^2$'s pull of $hh$ onto the $t\bar{t}$ shell.
- **Joint fit.** Fit angles and masses jointly rather than as a product, by adding Legendre-in-$u$ mass blocks to the extended block basis; with $m_{b\bar{b}}$ included the product form already fails because angles and masses become correlated.
- **Production variables.** $m_{t\bar{t}}$ and the $t\bar{t}$ transverse momentum are the remaining dimensions; a resonance such as $H \to t\bar{t}$ shows up in $m_{t\bar{t}}$.
- **Binning and defaults.** Equiprobable-contour binning of $t$; the truncated or extended $f$ as the default; an $f_{hh}$ for A–B testing.

## A Code and reproduction

Everything is in `/Users/TMHONG/Claude-code/ttbar_bblvqq` (git repository on `main`).

`setup_tools.sh` installs MadGraph5_aMC 3.6.7, Pythia8, Delphes 3.5.1 (built with CMake) and the Python environment.

`generate_samples.py` MadGraph → Pythia8 → Delphes production; `--process ttbar|hh|Htt`.

`plotting/ io.py` (LHE and Delphes readers), `physics.py` (objects, selection, $\chi^2$ reconstruction), `angles.py` (the ten angles), `osde.py` (basis functions, OSDE, leave-one-out, extended and spherical-harmonic bases), `summary.py`, `histo.py`, `style.py`.

`make_plots.py`, `plot_config.py` validation plots; `nu_pz_plot.py`, `lep_had_compare.py`, `hh_plots.py`, `acceptance_plots.py`.

`osde_fit.py` fits $f_{t\bar{t}}$ per level (`--basis paper|extended|sh`), applies it to $t\bar{t}$ and $hh$, projections and scores; `osde_simplify.py` (truncation, CV), `osde_order_scan.py`, `osde_truncation_check.py`, `osde_compare.py`, `basis_compare.py`, `paper_style_plots.py`.

`ml_compare.py` the unsupervised and supervised machine-learning comparison.

`study_washout/` the mechanism and the step-by-step decomposition of the parton-to-gen loss.

`study_salvage/` the mass extension (`simplest.py`, `salvage.py`, `reco_extension.py`), the angular-term study (`angle_terms.py`), the truncation-noise test (`truncation_noise.py`) and the tagging study (`tagging.py`).

`study_bsm/`, `pileup/` the $H \to t\bar{t}$ benchmark and the pileup study.

`make_slides.py`, `note/make_note.py` the slides and this note; both read the same result files.

`tests/test_osde.py` 14 unit tests of the basis functions and the estimator (`pytest`).

`LOG.md` the running log of the study.

The slides are rebuilt with `python make_slides.py --sample samples/ttbar_bblvqq/run_seed12345 --hh samples/hh_bblvqq/run_seed2001 --osde osde --acceptance acceptance` and this note with `python note/make_note.py`.

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